

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277570

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

OKUBO; Yasuhiro et al.

VEHICLE LAMP

Abstract

A vehicle lamp includes first and second light sources, a collecting lens that collects light, a shielding member that passes collected light through a radiating slit, and a projecting lens that forms a radiation pattern with light. The collecting lens includes a first lens portion and a second lens portion that correspond to the first and second light sources, respectively, the radiating slit includes a near-side slit portion and a far-side slit portion that correspond to a near-side radiation design and a far-side radiation design, respectively, in the radiation pattern, the first lens portion and the second lens portion are opposed to the far-side slit portion and the near-side slit portion, respectively. A collecting lens member, the shielding member, and a projecting lens member are fixed to a supporting member such that a collecting lens frame and a shielding frame overlap with each other in an optical axis direction.

Inventors: OKUBO; Yasuhiro (Isehara-shi, JP), KAWAMURA; Takayuki (Isehara-shi, JP)

Applicant: Ichikoh Industries, Ltd. (Isehara-shi, JP)

Family ID: 88519045

Assignee: Ichikoh Industries, Ltd. (Isehara-shi, JP)

Appl. No.: 18/858511

Filed (or PCT Filed): April 26, 2023

PCT No.: PCT/JP2023/016410

Foreign Application Priority Data

JP	2022-073748	Apr. 27, 2022
JP	2022-086939	May. 27, 2022
JP	2022-091900	Jun. 06, 2022

Publication Classification

Int. Cl.: F21S43/20 (20180101); F21W103/60 (20180101)

U.S. Cl.:

CPC F21S43/26411 (20240501); F21S43/26231 (20240501); B60Q2400/50 (20130101); F21W2103/60 (20180101)

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a vehicle lamp.

BACKGROUND ART

[0002] Vehicle lamps that form a radiation pattern on a road surface in the vicinity of a vehicle has been considered (for example, see Patent Literature 1). This conventional vehicle lamp projects light from a light source through a slit of a shade (shielding member) to form a radiation pattern and can inform a viewer of some intention. This conventional vehicle lamp guides the light from the light source to the shade with a light guide and thus efficiently uses the light from the light source. Further, a vehicle lamp described in Patent Literature 3 has a configuration in which the light from a light source is collected by a collecting lens, the collected light is emitted to a slit of a shielding member, and the light transmitted through the slit is emitted to a road surface in front of a vehicle by a projecting lens. With this configuration, each optical member of the collecting lens, the shielding member, and the projecting lens is supported by a supporting member in a state of being held by a component such as a frame or a case.

CITATION LIST

Patent Literature

[0003] PTL 1: Japanese Unexamined Patent Publication No. 2019-192350 [0004] PTL 2: Japanese Unexamined Patent Publication No. 2020-102332 [0005] PTL 3: Japanese Unexamined Patent Publication No. 2021-111465

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

[0006] However, in the conventional vehicle lamps, since the light from the light source is diffused in the light guide to make the light (light flux) distribution on the shade uniform, it is difficult to adjust the light distribution on the shade, and it is difficult to form a radiation pattern having a desired brightness distribution. Further, in the vehicle lamp described in Patent Literature 3, it is required to accurately attach each optical member.

[0007] The present disclosure has been made in view of the above-described circumstances, and an object of the present disclosure is to provide a vehicle lamp capable of forming a radiation pattern having a desired brightness distribution while efficiently using light from a light source, and capable of accurately attaching an optical component.

Means for Solving the Problem

[0008] A vehicle lamp according to the present disclosure includes a first light source and a second light source arranged in parallel, a collecting lens that collects light from the first light source and the second light source, a shielding member that is provided with a radiating slit that partially passes the light collected by the collecting lens, and a projecting lens that projects the light having passed through the shielding member to form a radiation pattern, the collecting lens includes a first lens portion that corresponds to the first light source and a second lens portion that corresponds to

the second light source, the radiating slit includes a near-side slit portion that corresponds to a near-side radiation design projected onto a near position in the radiation pattern and a far-side slit portion that corresponds to a far-side radiation design projected onto a farther position than the near-side radiation design in the radiation pattern, the first lens portion is provided to be opposed to the far-side slit portion, and the second lens portion is provided to be opposed to the near-side slit portion.

[0009] A vehicle lamp according to the present disclosure includes a plurality of light sources, a collecting lens that collects light from the plurality of light sources, a shielding member that is provided with a plurality of slit portions that partially passes the light collected by the collecting lens, and a projecting lens that projects the light having passed through the shielding member to form a radiation pattern including a plurality of radiation designs that corresponds to the plurality of slit portions, the light sources are provided individually corresponding to the slit portions, and the collecting lens includes a plurality of lens portions that individually corresponds to the slit portions and overlaps with each other and is provided with a light reduction portion that reduces light between the plurality of lens portions.

[0010] A vehicle lamp according to the present disclosure includes a light source, a collecting lens member in which a collecting lens that collects light emitted from the light source and a collecting lens frame holding the collecting lens are formed as one member, a shielding member in which a slit formation portion including a slit that partially passes a part of the light collected by the collecting lens and a shielding frame holding the slit formation portion are formed as one member, a projecting lens member provided with a projecting lens that projects the light having passed through the slit onto a road surface to form a radiation pattern, and a supporting member that supports the light source, and the collecting lens member, the shielding member, and the projecting lens member are fixed to the supporting member such that the collecting lens frame and the shielding frame overlap with each other in an optical axis direction.

[0011] In the above-described vehicle lamp, the entire collecting lens member is formed using a material forming the collecting lens, and the shielding member is formed to cover the collecting lens and the collecting lens frame of the collecting lens member when viewed from the front.

[0012] In the above-described vehicle lamp, in the projecting lens member, the projecting lens and a projecting lens frame holding the projecting lens are formed as one member, and the collecting lens member, the shielding member, and the projecting lens member are fixed to the supporting member such that the collecting lens frame, the shielding frame, and the projecting lens frame overlap with each other in the optical axis direction.

[0013] In the above-described vehicle lamp, the collecting lens of the collecting lens member is provided behind the collecting lens frame in the optical axis direction, and the projecting lens of the projecting lens member is provided in front of the projecting lens frame in the optical axis direction.

[0014] In the above-described vehicle lamp, the supporting member includes a base portion including a support surface that supports the light source and a fixing portion that fixes the collecting lens frame, the shielding frame, and the projecting lens frame, and the fixing portion protrudes from the support surface.

[0015] In the above-described vehicle lamp, the fixing portion is provided with an end surface serving as a reference surface to fix the collecting lens frame, the shielding frame, and the projecting lens frame at a leading end in a protruding direction, and the end surface is formed to be orthogonal to the optical axis direction.

Effect of the Invention

[0016] With the vehicle lamp according to the present disclosure, it is possible to form a radiation pattern having a desired brightness distribution while efficiently using light from a light source, and it is possible to accurately attach an optical component.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is an explanatory diagram illustrating a state in which a vehicle lamp according to a first embodiment of the present disclosure is mounted on a vehicle and forms each radiation pattern.

[0018] FIG. 2 is an explanatory diagram illustrating a configuration of the vehicle lamp.

[0019] FIG. 3 is an exploded view illustrating the configuration of the vehicle lamp.

[0020] FIG. 4 is an exploded view illustrating only optical members in the configuration of the vehicle lamp.

[0021] FIG. 5 is an explanatory diagram illustrating a configuration and a positional relationship of a first light source and a second light source.

[0022] FIG. 6 is an explanatory diagram illustrating a state of a collecting lens main body of a collecting lens as viewed from a shade side.

[0023] FIG. 7 is an explanatory diagram illustrating a positional relationship between the two light sources and the collecting lens main body of the collecting lens.

[0024] FIG. 8 is an explanatory diagram illustrating a light distribution in an inner ring light distribution area formed on the shade with light from the first light source, which is incident on a first lens portion of the collecting lens from a curved incidence surface portion and is emitted from an inner emission surface portion of a first emission surface.

[0025] FIG. 9 is an explanatory diagram illustrating a light distribution in an outer ring light distribution area formed on the shade with light from the first light source, which is incident on the first lens portion of the collecting lens from an annular incidence surface portion, is reflected by a reflective surface, and then is emitted from an outer emission surface portion of the first emission surface.

[0026] FIG. 10 is an explanatory diagram illustrating a light distribution of a first light distribution area formed by overlapping the inner ring light distribution area and the outer ring light distribution area by the first light source on the shade.

[0027] FIG. 11 is an explanatory diagram illustrating a light distribution in a second light distribution area formed on the shade with light from a second light source, which is incident on a second lens portion of the collecting lens from a second incidence surface and is emitted from a second emission surface.

[0028] FIG. 12 is an explanatory diagram illustrating a light distribution of a light distribution area on the shade by the first light source and the second light source.

[0029] FIG. 13 is an explanatory diagram illustrating a state (full lighting) in which a vehicle lamp according to a second embodiment of the present disclosure is mounted on a vehicle and forms each radiation pattern.

[0030] FIG. 14 is an exploded view illustrating a configuration of the vehicle lamp.

[0031] FIG. 15 is an explanatory diagram illustrating a configuration and a positional relationship of a first light source, a second light source, and a third light source in a light source portion.

[0032] FIG. 16 is an explanatory diagram illustrating a state of a collecting lens as viewed from the light source portion side.

[0033] FIG. 17 is an explanatory diagram illustrating a cross-section taken along the line I-I illustrated in FIG. 16.

[0034] FIG. 18 is an explanatory diagram illustrating a state of the collecting lens as viewed from the shade side.

[0035] FIG. 19 is an explanatory diagram illustrating a state of the collecting lens as viewed from the light source portion side and obliquely from a lower side in an up-down direction.

[0036] FIG. 20 is an explanatory diagram illustrating a state in which light from the second light

source passes between a first lens portion and a second lens portion to become stray light in the collecting lens.

[0037] FIG. **21** is an explanatory diagram illustrating a state in which stray light illustrated in FIG. **20** forms an unintended light distribution (stray light area) in a third slit portion when the light from the second light source having passed through the second lens portion forms a light distribution in a second slit portion of the shade.

[0038] FIG. **22** is an exploded perspective view illustrating an example of a vehicle lamp according to a third embodiment.

[0039] FIG. **23** is a diagram illustrating a state of the vehicle lamp as viewed from the front.

[0040] FIG. **24** is a diagram illustrating a configuration along a cross-section of F-F in FIG. **23**.

[0041] FIG. **25** is a diagram illustrating a configuration along a cross-section of G-G in FIG. **23**.

[0042] FIG. **26** is a diagram illustrating an example of an operation of the vehicle lamp.

[0043] FIG. **27** is a diagram illustrating an example of a radiation pattern formed on a road surface by the vehicle lamp.

MODE FOR CARRYING OUT THE INVENTION

[0044] A first embodiment of a vehicle lamp **10** as an example of a vehicle lamp according to the present disclosure will be described below with reference to the drawings. In FIG. **1**, in order to facilitate understanding of the state in which the vehicle lamp **10** is provided, the vehicle lamp **10** is illustrated in an emphasized manner with respect to a vehicle **1** and does not necessarily match the actual state. In addition, in FIG. **4**, optical members among the constituent members of the vehicle lamp **10**, i.e., the portions that optically act on light from both light sources (**21**, **22**) to form a radiation pattern P_i are extracted and illustrated. Further, in FIGS. **8** to **12**, only the periphery of a shade portion **51** (each of slit portions **54** of the shade portion **51**) is illustrated in order to facilitate understanding of the state in which each area is formed on a shade **15** (the slit portion **54** thereof). In FIGS. **8** to **12**, in each area illustrated, the area corresponding to the level of the light flux (light amount) is surrounded by a broken line, and the light (light flux) distribution is illustrated as a contour line in which the light flux increases toward the center of the area.

First Embodiment

[0045] The vehicle lamp **10** according to the first embodiment, which is an embodiment of the vehicle lamp according to the present disclosure, will be described with reference to FIGS. **1** to **12**. As illustrated in FIG. **1**, the vehicle lamp **10** according to the first embodiment is used as a lamp of the vehicle **1** such as an automobile and is provided in a front portion of the vehicle **1** to form the radiation pattern P_i on a road surface **2** in the vicinity of the front of the vehicle **1**, separately from a headlamp provided in the vehicle **1**. The vicinity of the front of the vehicle **1** always includes a proximity area closer to the vehicle **1** than a headlight area irradiated by the headlight provided in the vehicle **1** and may partially include the headlight area. The vehicle lamp **10** may also form the radiation pattern P_i on the road surface **2** in the vicinity of the rear or side of the vehicle **1** and is not limited to the configuration according to the first embodiment.

[0046] According to the first embodiment, each of the vehicle lamps **10** is provided on the front end of the vehicle **1** at a position higher than the road surface **2** and is provided in a state where a projection optical axis L_p (see FIG. **2** and the like) is inclined with respect to the road surface **2**. The two vehicle lamps **10** have basically the same configuration except that the mounting positions and the positions where the radiation patterns P_i are formed are different. In the description below, in each of the vehicle lamps **10**, the direction in which the projection optical axis L_p extends, i.e., the direction in which light is irradiated (projected), is an optical axis direction (Z in the drawings), a vertical direction when the optical axis direction extends along a horizontal plane is an up-down direction (Y in the drawings), and the direction (horizontal direction) orthogonal to the optical axis direction and the up-down direction is a width direction (X in the drawings) (see FIG. **2** and the like).

[0047] As illustrated in FIGS. **2** to **4**, in the vehicle lamp **10**, a light source portion **12**, a shielding

frame **13**, a collecting lens **14**, a shade **15**, and a projecting lens **16** are attached to an installation base portion **11** to form a single projection optical system and constitute a road surface projection unit of a projector type. The installation base portion **11** is a portion where the light source portion **12** is provided, is made of aluminum die-casting or resin having thermal conductivity, and functions as a heat sink that releases the heat generated in the light source portion **12** to the outside as a whole. The installation base portion **11** includes a base portion **21**, a plurality of heat radiation fins **22** (see FIG. 4), and a pair of mounting arm portions **23**.

[0048] The base portion **21** has a flat plate shape orthogonal to the optical axis direction and is provided with a light source mounting area **24** at the center thereof. The light source mounting area **24** defines an area where the light source portion **12** is mounted and, as illustrated in FIG. 3, is a flat surface and is provided with a pair of screw holes **24a** and a pair of positioning protrusions **24b**. The pair of screw holes **24a** is provided at diagonal positions in the light source mounting area **24** and can be fixed by screwing screws **25**. The pair of positioning protrusions **24b** is provided at ends of the light source mounting area **24** in a width direction and protrude forward in the optical axis direction. Each of the heat radiation fins **22** is formed in a plate shape orthogonal to the width direction on the rear side of the base portion **21** (the rear side in the optical axis direction (the side opposite to the direction in which light is emitted)). The heat radiation fins **22** are arranged side by side (in parallel) at predetermined intervals in the width direction.

[0049] In the base portion **21**, a pair of positioning protrusions **21a** is provided on the lower side of the two mounting arm portions **23**, and the screw hole **21b** is provided on the upper side of one of the mounting arm portions **23**. The positioning protrusion **21a** protrudes forward in the optical axis direction. The shielding frame **13** can be fixed by screwing a screw **26** into the screw hole **21b**. The installation base portion **11** radiates the heat generated in the light source portion **12** installed at the light source mounting area **24** to the outside mainly from each of the heat radiation fins **22**.

[0050] The pair of mounting arm portions **23** is provided as a pair on both outer sides in the width direction of the light source mounting area **24** and protrudes forward from the base portion **21** in the optical axis direction. End portions **23a** of the two mounting arm portions **23** on the front side in the optical axis direction are flat surfaces orthogonal to the optical axis direction, and the end portions **23a** are located at the same position (on the same plane) in the optical axis direction. Each of the end portions **23a** is provided with a positioning protrusion **23b** and a screw hole **23c**. The positioning protrusion **23b** is provided in a lower portion of the end portion **23a** in the up-down direction and protrudes forward in the optical axis direction. The screw hole **23c** is provided in an upper portion of the end portion **23a** in the up-down direction, and the collecting lens **14**, the shade **15**, and the projecting lens **16** can be fixed by screwing a screw **27**.

[0051] The light source portion **12** includes a first light source **31**, a second light source **32**, a connector terminal **33**, and a substrate **34** on which these components are mounted. The first light source **31** and the second light source **32** include a light emitting element such as a light emitting diode (LED). According to the first embodiment, the first light source **31** and the second light source **32** emit light in amber color (amber light) with a Lambertian distribution centered on the emission optical axis. The colors (wavelength bands), the mode of distribution, the number of colors, and the like, of the first light source **31** and the second light source **32** may be appropriately set and are not limited to the configuration according to the first embodiment.

[0052] As illustrated in FIG. 5 and the like, the first light source **31** and the second light source **32** according to the first embodiment are arranged in parallel in the up-down direction with the projection optical axis L_p interposed therebetween, the first light source **31** is located below the projection optical axis L_p , and the second light source **32** is located above the projection optical axis L_p . The first light source **31** has a rectangular shape elongated in the width direction and includes two LED chips **31a** and a phosphor **31b** covering each of them. The second light source **32** has a substantially square shape and includes one LED chip **32a** and a phosphor **32b** covering it. The first light source **31** and the second light source **32** emit light from the LED chips **31a**, **32a** as

amber light by passing the light through the phosphors **31b**, **32b**. Therefore, in the first light source **31**, the phosphor **31b** functions as a first light emission surface, and in the second light source **32**, the phosphor **32b** functions as a second light emission surface.

[0053] In the first light source **31** and the second light source **32**, emission optical axes **31L**, **32L** are set to extend from the respective centers in the optical axis direction. The emission optical axis **31L** and the emission optical axis **32L** are substantially parallel to the projection optical axis **Lp**. The emission optical axes **31L**, **32L** have a positional relationship such that the straight line connecting them passes through the projection optical axis **Lp**. According to the first embodiment, in the up-down direction, the distance from the projection optical axis **Lp** to the first light source **31** and the emission optical axis **31L** is smaller than the distance from the projection optical axis **Lp** to the second light source **32** and the emission optical axis **32L**. In other words, according to the first embodiment, there is a positional relationship such that the first light source **31** is closer to the projection optical axis **Lp** than the second light source **32**.

[0054] The connector terminal **33** is electrically connected to a wiring pattern of the substrate **34**, and a connection connector connected to a lighting control circuit is detachably attached. The connector terminal **33** is provided at a lower end portion of the substrate **34** in the up-down direction so that the connection connector can be easily attached and detached. When the connection connector is attached to the connector terminal **33**, electric power can be supplied from the lighting control circuit to the light sources **31**, **32** via the wiring pattern.

[0055] The substrate **34** has a plate shape formed of a resin material such as a glass epoxy substrate, and the first light source **31** and the second light source **32** are mounted on the substrate **34**. The substrate **34** is provided with a pair of screw through-holes corresponding to the pair of screw holes **24a** of the light source mounting area **24** of the base portion **21** of the installation base portion **11** and a pair of positioning holes **34a** corresponding to the pair of positioning protrusions **24b** of the light source mounting area **24**. The substrate **34** is attached to the light source mounting area **24** by inserting the corresponding positioning protrusions **24b** to the pair of positioning holes **34a** and screwing the screws **25** inserted into the pair of screw through-holes into the corresponding screw holes **24a**. As a result, the substrate **34** causes the light source mounting area **24**, i.e., the first light source **31** and the second light source **32** mounted thereon, to be opposed to the collecting lens **14**. The substrate **34** appropriately supplies power from the lighting control circuit via the connector terminal **33** to turn on the first light source **31** and the second light source **32**.

[0056] The shielding frame **13** prevents the light emitted from the first light source **31** and the second light source **32** from leaking out from between the light source portion **12** (the substrate **34**) and the collecting lens **14** and includes a frame portion main body **35** and a mounting portion **36**. The frame portion main body **35** includes a plate-like member that has a predetermined dimension in the optical axis direction and extends in an annular shape to surround substantially the entire area of the light source portion **12** (the substrate **34**). In the frame portion main body **35**, the dimension in the optical axis direction is partially reduced on the lower side in the up-down direction, and the connector terminal **33** of the light source portion **12** can be attached to and detached from the connection connector. The pair of mounting portions **36** has a plate shape protruding from the frame portion main body **35** in the width direction at positions corresponding to the positioning protrusion **21a** and the screw hole **21b** of the base portion **21**. The mounting portion **36** corresponding to the positioning protrusion **21a** is provided with a positioning hole **36a**, and the mounting portion **36** corresponding to the screw hole **21b** is provided with a screw through-hole through which the screw **26** can be inserted. The shielding frame **13** is attached to the base portion **21** of the installation base portion **11** by inserting the positioning protrusion **21a** into the positioning hole **36a** and screwing the screw **26** inserted into the screw through-hole into the screw hole **21b**.

[0057] The collecting lens **14** collects the light emitted from the first light source **31** and the second

light source **32** and collects the light in the periphery of each of the slit portions **54** described below on the shade **15**, i.e., in the area where each of the slit portions **54** is provided while including all of the slit portions **54** on the shade **15**. As illustrated in FIG. 3, the collecting lens **14** includes a collecting lens main body **41** that collects light from the first light source **31** and the second light source **32** and a pair of collecting lens mounting pieces **42** that protrudes from the collecting lens main body **41** in the width direction. The collecting lens main body **41** is basically a convex lens, and optical characteristics thereof are set to form an inner ring light distribution area A_i and an outer ring light distribution area A_o (see FIGS. 8 and 9) on the shade **15**. This will be described below.

[0058] The two collecting lens mounting pieces **42** are formed in a plate shape orthogonal to the optical axis direction and can be applied to the end portions **23a** of the two mounting arm portions **23** of the base portion **21** of the installation base portion **11**. Each of the collecting lens mounting pieces **42** is provided with a collecting lens positioning hole **42a** and a collecting lens screw through-hole **42b**. Each of the collecting lens positioning holes **42a** allows the positioning protrusion **23b** to be fitted into the collecting lens positioning hole **42a** in a state where the collecting lens mounting piece **42** is applied to the end portion **23a**. Each of the collecting lens screw through-holes **42b** allows the screw **27** screwed into the screw hole **23c** to be inserted into the collecting lens screw through-holes **42b** in a state where the collecting lens mounting piece **42** is applied to the end portion **23a**. The collecting lens **14** is attached to the two mounting arm portions **23** (the end portions **23a** thereof) of the installation base portion **11** by inserting the corresponding positioning protrusions **23b** into the respective collecting lens positioning holes **42a** and screwing the respective screws **27** inserted into the respective collecting lens screw through-holes **42b** into the corresponding screw holes **23c**.

[0059] The shade **15** is an example of a shielding member that forms the radiation pattern P_i by causing the light from the first light source **31** or the second light source **32** collected by the collecting lens **14** to partially pass through a radiating slit **53**. As illustrated in FIG. 1, in the radiation pattern P_i , three radiation designs D_i are arranged at substantially equal intervals in a direction away from the vehicle **1**. In this case, when each of the radiation designs D_i is individually illustrated, the radiation design farthest from the vehicle **1** is defined as a first radiation design D_{i1} , and the radiation designs are defined as a second radiation design D_{i2} and a third radiation design D_{i3} in order as they are closer to the vehicle **1**. According to the first embodiment, the first radiation design D_{i1} and the second radiation design D_{i2} are V-shaped symbols that are widely opened, and the first radiation design D_{i1} is larger than the second radiation design D_{i2} . The third radiation design D_{i3} has an elongated pentagonal shape extending from the vehicle **1** toward the second radiation design D_{i2} , a base end on the vehicle **1** side has a shape in which corners are provided at both ends of a straight line orthogonal to an arrow direction D_a described below, and the other end away from the vehicle **1** has a V shape corresponding to the first radiation design D_{i1} and the second radiation design D_{i2} . The dimension of the third radiation design D_{i3} in the direction orthogonal to the arrow direction D_a is smaller than the first radiation design D_{i1} and the second radiation design D_{i2} . In the radiation pattern P_i , the first radiation design D_{i1} and the second radiation design D_{i2} are far-side radiation designs, and the third radiation design D_{i3} is a near-side radiation design.

[0060] The radiation pattern P_i is formed by arranging the first radiation design D_{i1} , the second radiation design D_{i2} , and the third radiation design D_{i3} on the same straight line so as to be away from the vehicle **1** on the road surface **2** as a projection surface. Therefore, the radiation pattern P_i can look like an arrow pointing in the arrow direction D_a from the vehicle **1** by arranging the three radiation designs D_i . The direction indicated by the arrow as the radiation pattern P_i , i.e., the direction in which the apexes of the V shapes of the radiation designs D_i are arranged, is defined as the arrow direction D_a , and the side indicated by the arrow direction D_a (the first radiation design D_{i1} side) is defined as the front side of the arrow direction D_a . According to the first embodiment,

the three radiation designs D_i have both side ends D_{ie} located in a direction orthogonal to the arrow direction D_a . The both side ends D_{ie} are located on the same straight line that is inclined toward the inner side (the side closer to the vehicle **1**) as it is closer to the back side in the arrow direction D_a and emphasize an impression that the three radiation designs D_i constitute the arrow indicating the arrow direction D_a . The radiation pattern P_i including the three radiation designs D_i is formed by the shade **15**.

[0061] As illustrated in FIGS. **3** and **4**, the shade **15** is made of a plate-like member that basically blocks transmission of light and includes a shade portion **51** and a pair of shade mounting pieces **52**. The shade mounting pieces **52** protrude from both sides of the shade portion **51** in the width direction and can be applied to the respective collecting lens mounting pieces **42** of the collecting lens **14** attached to the end portions **23a** of the two mounting arm portions **23** of the installation base portion **11**. Each of the shade mounting pieces **52** is provided with a shade positioning hole **52a** and a shade screw through-hole **52b**. Each of the shade positioning holes **52a** allows the positioning protrusion **23b** passed therethrough to be fitted into the shade positioning hole **52a** in a state where the shade mounting piece **52** is applied to the collecting lens mounting piece **42**. Each of the shade screw through-holes **52b** allows the screw **27** inserted into the collecting lens screw through-hole **42b** to be inserted into the shade screw through-hole **52b** in a state where the shade mounting piece **52** is applied to the collecting lens mounting piece **42**. The shade **15** is attached to the two mounting arm portions **23** of the installation base portion **11** via the collecting lens **14** by inserting the corresponding positioning protrusion **23b** to each of the shade positioning holes **52a** and screwing each of the screws **27** inserted into each of the shade screw through-holes **52b** into the corresponding screw hole **23c**. In the shade **15**, the center position of the shade portion **51** is located on the projection optical axis L_p by attaching the shade mounting piece **52** to the two mounting arm portions **23**.

[0062] The shade portion **51** is provided with the radiating slit **53** formed by partially cutting out and penetrating the plate-like member. The radiating slit **53** partially transmits the light from the first light source **31** and the second light source **32** collected by the collecting lens **14** (the collecting lens main body **41**), thereby forming the radiation pattern P_i to be projected into a predetermined shape. The radiating slit **53** corresponds to the radiation pattern P_i and, as illustrated in FIGS. **3**, **4**, **8**, and the like, includes the three slit portions **54** according to the first embodiment.

[0063] The three slit portions **54** correspond one to-one to the three radiation designs D_i . Since the projecting lens **16** inverts the shade **15** (the radiating slit **53**) and projects the inverted shade **15** onto the road surface **2**, each of the slit portions **54** has a rotationally symmetrical positional relationship around the projection optical axis L_p with respect to the positional relationship of each of the radiation designs D_i of the radiation pattern P_i (see FIGS. **3** and **4**). Therefore, in each of the slit portions **54**, a first slit portion **541** on the lowermost side in the up-down direction corresponds to the first radiation design D_{i1} of the radiation pattern P_i , and a second slit portion **542** above it corresponds to the second radiation design D_{i2} . Thus, the first slit portion **541** and the second slit portion **542** are far-side slit portions corresponding to the first radiation design D_{i1} and the second radiation design D_{i2} which are far-side radiation designs. In each of the slit portions **54**, a third slit portion **543** on the uppermost side is a near-side slit portion corresponding to the third radiation design D_{i3} , i.e., a near-side radiation design.

[0064] The position of each of the slit portions **54** on the shade portion **51** is set such that each of the radiation designs D_i has a target size and a target positional relationship on the road surface **2**. In the shade **15** according to the first embodiment, the third slit portion **543** is provided above the projection optical axis L_p in the up-down direction, the second slit portion **542** is provided below it, and the first slit portion **541** is provided below it. The substantially entire second slit portion **542** is located below the horizontal line including the projection optical axis L_p , and parts of both end portions in the width direction are located above the horizontal line including the projection optical axis L_p so as to across the horizontal line. Therefore, in the shade **15** (the shade portion **51**), three

quarters or more of the far-side slit portions (the first slit portion **541** and the second slit portion **542**) are provided below the projection optical axis L_p of the projecting lens **16**. The light transmitted through the shade **15** (each of the slit portions **54** of the radiating slit **53**) is projected onto the road surface **2** by the projecting lens **16**.

[0065] The first slit portion **541** and the second slit portion **542**, which are far-side slit portions, are formed in a shape resembling a largely opened V-shaped symbol similarly to the respective corresponding radiation designs D_i and are vertically and horizontally inverted with respect to the respective radiation designs D_i . Further, the third slit portion **543**, which is a near-side slit portion, is formed in a pentagonal shape extending from the upper side toward the second slit portion **542** similarly to the corresponding third radiation design D_{i3} and is vertically and horizontally inverted with respect to the third radiation design D_{i3} . The sizes and the intervals of the three slit portions **54** are set in accordance with the distances to the road surface **2** so that the radiation designs D_i have the sizes illustrated in FIG. **1** and are arranged at substantially equal intervals on the road surface **2**. Specifically, in the vehicle lamp **10**, since the projection optical axis L_p is provided to be inclined with respect to the road surface **2**, the distances from the shade **15** and the projecting lens **16** to the road surface **2** are different, and therefore, when projected onto the road surface **2** by the projecting lens **16**, each of the slit portions **54** (each of the radiation designs D_i that is light transmitted therethrough) has the size and the interval corresponding to the distance.

[0066] Therefore, the size and the interval of each of the slit portions **54** are set in accordance with the distance to the road surface **2** so that the radiation designs D_i have the above-described size and are arranged at substantially equal intervals on the road surface **2**. Specifically, according to the first embodiment, the first slit portion **541** has a shape resembling a thin V-shaped symbol, and the second slit portion **542** has a shape resembling a V-shaped symbol that is thicker than the first slit portion **541**. In the shade portion **51**, the second slit portion **542** has a positional relationship in which a bent portion of an upper outline included in the outline of the V-shaped symbol protruding downward coincides with the projection optical axis L_p (see FIG. **8** and the like). Further, the third slit portion **543** has a shape similar to a regular pentagon shape protruding toward the second slit portion **542**. Each of the slit portions **54** is formed in a shape wider than the corresponding radiation design D_i in the width direction.

[0067] In this manner, the three slit portions **54** are different in size from the respective radiation designs D_i and are arranged at different intervals. In each of the slit portions **54**, the first slit portion **541** has the smallest reduction ratio with respect to the corresponding radiation design D_i , and the passed light is enlarged at the largest enlargement ratio when projected on the road surface **2** to form the first radiation design D_{i1} . In addition, in each of the slit portions **54**, the third slit portion **543** has the largest reduction ratio with respect to the corresponding radiation design D_i , and the transmitted light is enlarged at the smallest enlargement ratio when projected onto the road surface **2** to form the third radiation design D_{i3} .

[0068] As illustrated in FIGS. **2** and **3**, the projecting lens **16** includes a projecting lens main body **55** that projects light having passed through the shade **15** and a pair of projecting lens mounting pieces **56** that protrudes from it in the width direction. As illustrated in FIG. **4** and the like, the projecting lens main body **55** is a convex lens having a circular shape when viewed in the optical axis direction and is a free-form surface in which the incidence surface and the emission surface are convex surfaces according to the first embodiment. The projecting lens main body **55** forms the radiation pattern P_i on the road surface **2** inclined with respect to the projection optical axis L_p as illustrated in FIG. **1** by projecting the radiating slit **53** (each of the slit portions **54** thereof) of the shade **15**. The incidence surface and the emission surface may be convex or concave surfaces as long as the projecting lens main body **55** is a convex lens and are not limited to the configuration according to the first embodiment.

[0069] The two projecting lens mounting pieces **56** are formed in a plate shape orthogonal to the optical axis direction and can be applied to the shade mounting pieces **52** of the shade **15** attached

to the end portions **23a** of the two mounting arm portions **23** of the installation base portion **11**. Each of the projecting lens mounting pieces **56** is provided with a projecting lens positioning hole **56a** and a projecting lens screw through-hole. Each of the projecting lens positioning holes **56a** allows the positioning protrusion **23b** passed therethrough to be fitted into the projecting lens positioning hole **56a** in a state where the projecting lens mounting piece **56** is applied to the shade mounting piece **52**. Each of the projecting lens screw through-holes allows the screw **27** inserted through the shade screw through-hole **52b** to be inserted into the projecting lens screw through-hole in a state where the projecting lens mounting piece **56** is applied to the shade mounting piece **52**. The projecting lens **16** is attached to the two mounting arm portions **23** (the end portions **23a** thereof) of the installation base portion **11** by inserting the corresponding positioning protrusion **23b** into each of the projecting lens positioning holes **56a** and screwing each of the screws **27** inserted into the respective projecting lens screw through-holes into the corresponding screw hole **23c**. Accordingly, in the projecting lens **16**, the projection optical axis L_p , which is the optical axis of the projecting lens main body **55**, is directed in a predetermined direction, and the direction of the projection optical axis L_p of the vehicle lamp **10** is set.

[0070] Next, the configuration of the collecting lens main body **41** of the collecting lens **14** will be described mainly with reference to FIGS. **6** and **7**. The collecting lens main body **41** includes a first lens portion **61** that corresponds to the first light source **31** and a second lens portion **62** that corresponds to the second light source **32**. In the collecting lens **14** (the collecting lens main body **41**) according to the first embodiment, the first lens portion **61** and the second lens portion **62** are integrally formed in a state where the second lens portion **62** is placed on the first lens portion **61**.

[0071] The first lens portion **61** has a substantially circular shape in a front view in the optical axis direction. The first lens portion **61** according to the first embodiment collects the light emitted from the first light source **31** and having a spread as a whole in a state close to being parallel to the projection optical axis L_p and causes the light to travel to the shade portion **51**. The first lens portion **61** includes a first incidence surface **63** opposed to the first light source **31** and a first emission surface **64** directed to the opposite side.

[0072] As illustrated in FIG. **7**, a central portion of the first incidence surface **63** is recessed to the inside of the first lens portion **61** (the side opposite to the first light source **31**), and the first incidence surface **63** includes a curved incidence surface portion **65** curved convexly outward at the center thereof and an annular incidence surface portion **66** surrounding it. A truncated cone-shaped reflective surface **67** surrounding the annular incidence surface portion **66** is provided around the first incidence surface **63**.

[0073] The curved incidence surface portion **65** is opposed to the first light source **31** in the optical axis direction, and the first light source **31** is located in the vicinity of the focal point on the rear side (rear focal point). The curved incidence surface portion **65** causes light emitted from the first light source **31** to be incident on the first lens portion **61** as parallel light traveling substantially parallel to the axis line of the first lens portion **61**. This parallel light (light that is parallel) refers to light that is collimated when the light passes through the curved incidence surface portion **65**.

[0074] The annular incidence surface portion **66** is provided to protrude from the curved incidence surface portion **65** toward the first light source **31** and causes light that does not travel to the curved incidence surface portion **65** among the light from the first light source **31** to be incident on the first lens portion **61**. The reflective surface **67** is formed at a position where the light entering the first lens portion **61** from the annular incidence surface portion **66** travels. When the reflective surface **67** reflects the light incident from the annular incidence surface portion **66**, the light becomes parallel light traveling substantially parallel to the axis line of the first lens portion **61**. The reflective surface **67** may reflect light by using total reflection, or may reflect light by adhering aluminum, silver, or the like by vapor deposition, coating, or the like. Accordingly, the first incidence surface **63** causes the light emitted from the first light source **31** to travel into the first lens portion **61** as parallel light traveling substantially parallel to the axis line of the first lens

portion **61**, and guides the light to the first emission surface **64**.

[0075] Therefore, in the first lens portion **61**, on the first incidence surface **63**, the light having passed through the curved incidence surface portion **65** becomes direct light that directly travels toward the first emission surface **64**, and the light having passed through the annular incidence surface portion **66** and reflected by the reflective surface **67** becomes reflected light that travels toward the first emission surface **64** after being internally reflected.

[0076] The first emission surface **64** emits light, which is incident from the first incidence surface **63** and is converted into parallel light, to the front side in the front-back direction. As illustrated in FIG. **6** and the like, the first emission surface **64** has a substantially circular shape in a front view and includes an inner emission surface portion **68** and an outer emission surface portion **69** having different optical settings. The inner emission surface portion **68** is provided in an area which is in the vicinity of the center of the first emission surface **64** and in which light having passed through the curved incidence surface portion **65** travels. The inner emission surface portion **68** according to the first embodiment has a substantially circular shape in a front view. The inner emission surface portion **68** refracts the light having passed through the curved incidence surface portion **65** to cause the light to travel toward the front side in the front-back direction while greatly diffusing the light in the width direction (horizontal direction). The inner emission surface portion **68** forms a plurality of light distribution images of the first light source **31** at positions corresponding to the optical characteristics on the shade **15** (the shade portion **51**) in an appropriately overlapping manner by emitting light from the first light source **31** through the curved incidence surface portion **65**. The optical characteristics can be set by adjusting the curvatures (surface shapes) of the inner emission surface portion **68** together with the curved incidence surface portion **65** for each place and, according to the first embodiment, the curvatures are set to be gradually changed.

[0077] The inner emission surface portion **68** appropriately refracts the light emitted from the first light source **31** and passing through the curved incidence surface portion **65**, thereby irradiating the light onto the shade **15** to form the inner ring light distribution area A_i illustrated in FIG. **8**. In the inner ring light distribution area A_i , the entire area of the first slit portion **541** corresponding to the far side in the far-side slit portion (forming the first radiation design $Di1$ located on the far side in the far-side radiation design) has a substantially equal light amount on the shade **15**.

[0078] As illustrated in FIG. **6** and the like, the outer emission surface portion **69** is provided in an area surrounding the inner emission surface portion **68** and is located in an area where the light emitted from the first light source **31** and reflected by the reflective surface **67** via the annular incidence surface portion **66** travels. The outer emission surface portion **69** refracts the light emitted from the first light source **31**, passing through the annular incidence surface portion **66**, and reflected by the reflective surface **67**, thereby causing the light to travel toward the front side in the front-back direction to be collected on the axis line side of the first lens portion **61**. The outer emission surface portion **69** emits the light reflected by the reflective surface **67** to form a plurality of light distribution images of the first light source **31** at positions corresponding to the optical characteristics on the shade **15** in an appropriately overlapping manner. The optical characteristics can be set by adjusting the curvatures (surface shapes) of the outer emission surface portion **69** together with the reflective surface **67** for each place and, according to the first embodiment, the curvatures are set to be gradually changed.

[0079] The outer emission surface portion **69** appropriately refracts the light which is emitted from the first light source **31**, passes through the annular incidence surface portion **66**, and is reflected by the reflective surface **67**, thereby irradiating the light onto the shade **15** to form the outer ring light distribution area A_o illustrated in FIG. **9**. The outer ring light distribution area A_o is an area where a high light flux (light amount) is formed on the shade **15**. The outer ring light distribution area A_o has the maximum value of the light flux from the apex of the first slit portion **541** to the middle position in the width direction and in the vicinity of the apex of the second slit portion **542** and surrounds the entire area of the remaining far-side slit portions, i.e., the entire remaining area of the

first slit portion **541** and the second slit portion **542**, with a light flux lower than the maximum value. In the outer ring light distribution area **Ao**, there is a large difference in light flux between the area where the light flux has the maximum value and the area surrounding it, and the change in light flux at the boundary is continuous. In the outer ring light distribution area **Ao**, the central light flux having the maximum value is at least higher than that in the inner ring light distribution area **Ai**. Here, since the lens optical axis of the first lens portion **61** extends toward the apex of the first slit portion **541**, it is easy to form the outer ring light distribution area **Ao** as described above.

[0080] In this way, the first lens portion **61** forms the inner ring light distribution area **Ai** to irradiate the entire area of the first slit portion **541** with the light from the first light source **31** via the inner emission surface portion **68** and forms the outer ring light distribution area **Ao** to irradiate the entire area of the far-side slit portion while providing the highest light flux in the vicinity of the apex of the first slit portion **541** with the light from the first light source **31** via the outer emission surface portion **69**. Therefore, the first lens portion **61** forms a first light distribution area **A1** illustrated in FIG. **10** by causing the light from the first light source **31**, i.e., the direct light traveling toward the inner emission surface portion **68** via the curved incidence surface portion **65** and the reflection light reflected by the reflective surface **67** via the annular incidence surface portion **66** and traveling toward the outer emission surface portion **69**, to pass through different optical paths as described above. The first light distribution area **A1** is formed by overlapping the inner ring light distribution area **Ai** and the outer ring light distribution area **Ao** described above and irradiates the first slit portion **541** and the second slit portion **542**, i.e., the entire far-side slit portion, with the highest light flux from the apex of the first slit portion **541** to the middle position in the width direction and in the vicinity of the apex of the second slit portion **542**.

[0081] As illustrated in FIGS. **6** and **7**, the second lens portion **62** is a convex lens having a substantially rectangular shape elongated in the width direction when viewed from the front in the optical axis direction and, as a whole, collects the light emitted from the second light source **32** and having a spread in a state close to being parallel to the projection optical axis **Lp** and causes the light to travel to the shade portion **51**. The second lens portion **62** includes a second incidence surface **71** opposed to the second light source **32** and a second emission surface **72** directed to the opposite side. Therefore, the second incidence surface **71** forms an incidence surface of the collecting lens **14** together with the first incidence surface **63** of the first lens portion **61**. In addition, the second emission surface **72** forms an emission surface of the collecting lens **14** together with the first emission surface **64** of the first lens portion **61**. According to the first embodiment, the second lens portion **62** is a free-form surface in which the second incidence surface **71** and the second emission surface **72** are convex surfaces. The second incidence surface **71** and the second emission surface **72** may be convex surfaces or concave surfaces as long as the second lens portion **62** is a convex lens, and are not limited to the configuration according to the first embodiment.

[0082] The second incidence surface **71** is opposed to the second light source **32** in the optical axis direction, and the second light source **32** is located in the vicinity of the focal point on the rear side (rear focal point). The curved incidence surface portion **65** causes light emitted from the second light source **32** to be incident on the second lens portion **62** as parallel light traveling substantially parallel to the axis line of the second lens portion **62**. The second emission surface **72** is provided on the side opposite to the second incidence surface **71** and refracts the light having passed through the second incidence surface **71** to cause the light to travel toward the front side in the front-back direction while diffusing the light. The second emission surface **72** forms a plurality of light distribution images of the second light source **32** at positions corresponding to the optical characteristics on the shade **15** (the shade portion **51**) in an appropriately overlapping manner by emitting light from the second light source **32** via the second incidence surface **71**. The optical characteristics can be set by adjusting the curvatures (surface shapes) of the second emission surface **72** together with the second incidence surface **71** for each place and, according to the first

embodiment, the curvatures are set to be gradually changed.

[0083] The second emission surface **72** appropriately refracts the light emitted from the second light source **32** and passing through the second incidence surface **71**, thereby irradiating the light onto the shade **15** to form a second light distribution area **A2** illustrated in FIG. **11**. In the second light distribution area **A2**, the entire area of the third slit portion **543**, which is a near-side slit portion, has a substantially equal light flux (light amount) on the shade **15**. Here, a substantially equal light flux in the entire area means that a change in the light flux is smaller than that in at least the outer ring light distribution area **Ao**, and preferably means a substantially uniform light flux. It is assumed that the second light distribution area **A2** according to the first embodiment has a lower light flux than the first light distribution area **A1**, i.e., the inner ring light distribution area **Ai** and the outer ring light distribution area **Ao**.

[0084] Accordingly, the collecting lens **14** forms the light distribution area **A** illustrated in FIG. **12** by overlapping the first light distribution area **A1** formed by the first lens portion **61** with the light from the first light source **31** and the second light distribution area **A2** formed by the second lens portion **62** with the light from the second light source **32**. The light distribution area **A** provides the highest light flux from the apex of the first slit portion **541** to the middle position in the width direction and in the vicinity of the apex of the second slit portion **542**, has the next highest light flux in the remaining portions of the first slit portion **541** and the second slit portion **542**, and is uniform in the entire area of the third slit portion **543** with a light flux lower than them. For this reason, the collecting lens **14** according to the first embodiment sets the vicinity of the apex of the first slit portion **541** forming the first radiation design **Di1** as the maximum light flux place on the shade **15**. As described above, on the shade **15**, the collecting lens **14** can form a predetermined light flux distribution with a predetermined modulation (inflection (differences in the level of the light flux)) for the far-side slit portion and can form a uniform light flux lower than that of the far-side slit portion for the near-side slit portion.

[0085] Next, the operation of the vehicle lamp **10** will be described. The vehicle lamp **10** supplies power from the lighting control circuit to the two light sources (**21**, **22**) from the substrate **34** and thus can appropriately turn on and off them. The light from the two light sources (**21**, **22**) is collected by the collecting lens **14** to irradiate the shade **15**, passes through the radiating slit **53** (each of the slit portions **54**), and is then projected by the projecting lens **16**, thereby forming the radiation pattern **Pi** on the road surface **2**. In the radiation pattern **Pi**, the light transmitted through the radiating slit **53** (each of the slit portions **54**) of the shade **15** having the light (light flux) distribution described above is projected by the projecting lens **16**, and thus the three radiation designs **Di** are arranged substantially on the straight line while the first radiation design **Di1**, in particular, the vicinity of the leading end thereof has the highest light flux. Further, in the vehicle lamp **10** according to the first embodiment, since the first light source **31** and the second light source **32** emit monochromatic light, it is possible to significantly suppress the influence of chromatic aberration in the projecting lens **16**, and it is possible to make the radiation pattern **Pi**, i.e., each of the radiation designs **Di** clear.

[0086] The vehicle lamp **10** is interlocked with the turn lamps, and when one of the right and left turn lamps is turned on, the first light source **31** and the second light source **32** provided on the turned-on side are turned on to form the radiation pattern **Pi** on the road surface **2**. Therefore, in a scene where the vehicle **1** is traveling from an alley with poor visibility to a different alley, the vehicle lamp **10** can cause a person on the different alley to visually recognize the radiation pattern **Pi** formed on the road surface **2** even when the person cannot visually recognize the vehicle **1**. In addition, in the vehicle **1**, when the hazard lamps are turned on, the two left and right vehicle lamps **10** simultaneously form the radiation patterns **Pi** on the road surface **2**, and therefore, it is possible to more reliably recognize that the hazard lamps are turned on, compared to a case where only the left and right turn lamps are blinked.

[0087] Further, in the radiation pattern **Pi** formed by the vehicle lamp **10**, the first radiation design

Di1 and the second radiation design Di2, which are far-side radiation designs, are formed in a V shape, and the third radiation design Di3, which is a near-side radiation design, is formed in a pentagonal shape elongated in the arrow direction Da. Therefore, the vehicle lamp **10** can give an impression that the far-side radiation design (the first radiation design Di1 and the second radiation design Di2) corresponds to the arrow head in the symbol of the arrow and the near-side radiation design (the third radiation design Di3) corresponds to the shaft in the symbol of the arrow, and can more effectively give an impression of pointing in the arrow direction Da. In addition, in the radiation pattern Pi according to the first embodiment, the both side ends Die of the three radiation designs Di are located on the same straight line inclined toward the inner side (the side closer to the vehicle **1**) as it is closer to the back side in the arrow direction Da. Accordingly, the vehicle lamp **10** can cause a person around the vehicle **1** to more intuitively recognize that the radiation pattern Pi indicates the arrow direction Da.

[0088] In the vehicle lamp **10**, the first light source **31** of the light source portion **12** is configured such that the two LED chips **31a** are arranged in parallel in the width direction, and the corresponding far-side slit portions, i.e., the first slit portion **541** and the second slit portion **542**, are formed in a shape that resembles a V-shaped symbol that is elongated in the width direction. Therefore, the vehicle lamp **10** can efficiently irradiate the far-side slit portion over the entire area in the width direction, and the far-side radiation design formed by the far-side slit portion, i.e., the first radiation design Di1 and the second radiation design Di2, can be brightened over the entire area.

[0089] Here, in the conventional vehicle lamp described in the related art document, a plurality of light guides is provided individually corresponding to a plurality of light sources, and light from each light source is efficiently used. In the conventional vehicle lamp, each of the light guide members diffuses light therein to emit light having a substantially uniform light flux distribution, and a shade (shielding member) is irradiated with the light passing through each of the light guide members to obtain a substantially uniform light (light flux) distribution on the shade. Further, in the conventional vehicle lamp, since the light from the light source corresponding to each light guide is guided onto the shade, the light is separately guided onto the shade for each light source. For this reason, it is difficult for the conventional vehicle lamp to obtain a desired light distribution as in the case where an area having a high light flux is partially formed while continuously changing the light flux in an area having a low light flux on the shade.

[0090] On the other hand, for the two light sources (**21**, **22**), the vehicle lamp **10** is provided with the single collecting lens **14** that collects light from each of them. The collecting lens **14** includes the first lens portion **61** that corresponds to the first light source **31** and the second lens portion **62** that corresponds to the second light source **32**. In the first lens portion **61**, the light emitted from the first light source **31** in a direction substantially along the emission optical axis **31L** is incident from the curved incidence surface portion **65** of the first incidence surface **63**, and the light emitted from the first light source **31** in a spreading direction (at a large angle with respect to the emission optical axis **31L**) is incident from the annular incidence surface portion **66** of the first incidence surface **63** and reflected by the reflective surface **67**. The first lens portion **61** emits the light passing through the curved incidence surface portion **65** mainly from the inner emission surface portion **68** of the first emission surface **64** and emits the light passing through the annular incidence surface portion **66** and reflected by the reflective surface **67** mainly from the outer emission surface portion **69** of the first emission surface **64**. Then, the first lens portion **61** forms the inner ring light distribution area Ai on the shade **15** with the light having passed through the curved incidence surface portion **65** and the inner emission surface portion **68** and forms the inner ring light distribution area Ai on the shade **15** with the light having passed through the annular incidence surface portion **66**, reflected by the reflective surface **67**, and passed through the outer emission surface portion **69**. In this way, in the first lens portion **61**, the outer ring light distribution area Ao and the inner ring light distribution area Ai having different positions, sizes, and light (light flux)

distributions are formed using the difference between the optical path passing through the curved incidence surface portion **65** and the inner emission surface portion **68** and the optical path passing through the annular incidence surface portion **66**, the reflective surface **67**, and the outer emission surface portion **69**. The first lens portion **61** overlaps the outer ring light distribution area A_o and the inner ring light distribution area A_i having different light distributions, thereby forming the first light distribution area $A1$ having a light flux distribution with a predetermined modulation (inflection (differences in the level of the light flux)) on the far-side slit portion (the first slit portion **541** and the second slit portion **542**).

[0091] In addition, since the second lens portion **62** is a convex lens, the light from the second light source **32** is incident from the second incidence surface **71** and is emitted from the second emission surface **72**, and the second light distribution area $A2$, which is lower than the first light distribution area $A1$ and has a uniform light flux, is formed on the near-side slit portion (the third slit portion **543**) of the shade **15**. Here, the near-side slit portion corresponds to the third radiation design $Di3$, which is a near-side radiation design in the radiation pattern P_i and is projected onto a place close to the vehicle **1**, i.e., a place at a close distance from the vehicle lamp **10**, and therefore, in order to form the near-side radiation design which is easy to see, it is desirable that the light flux be uniform and lower than that of the far-side slit portion.

[0092] Therefore, by using the single collecting lens **14** for both the light sources (**21**, **22**), the vehicle lamp **10** can emit light onto the shade **15** with a desired light distribution and can form the first light distribution area $A1$ having a desired light flux distribution and the second light distribution area $A2$ having a uniform light flux lower than the first light distribution area $A1$. Then, the vehicle lamp **10** forms the light distribution area A by overlapping the first light distribution area $A1$ and the second light distribution area $A2$ on the shade **15**, thereby forming the radiation pattern P_i that is easy to see with a desired brightness distribution. Therefore, as compared with conventional vehicle lamps, the vehicle lamp **10** can easily adjust the light distribution of the light distribution area A formed on the shade **15** and can form the radiation pattern P_i having a desired brightness distribution while having a simple configuration using the single collecting lens **14**.

[0093] Here, the third radiation design $Di3$ according to the first embodiment has a shape elongated in the arrow direction Da and extends to the vicinity of the vehicle **1** in which the vehicle lamp **10** is mounted. In other words, the third radiation design $Di3$ according to the first embodiment is formed in a shape elongated in the arrow direction Da to reduce the interval with the vehicle **1**. Then, since the third slit portion **543** corresponds to the third radiation design $Di3$, the third slit portion **543** has a large dimension on the shade portion **51**. Then, when the third slit portion **543** is irradiated by using a lens having a predetermined light flux distribution in which the light flux is modulated as in the first lens portion **61**, it is difficult to irradiate the entire area, and unevenness of the brightness, such as the occurrence of a dark place, occurs in the formed third radiation design $Di3$, and there is a concern that the appropriate third radiation design $Di3$ cannot be formed.

[0094] In contrast, the vehicle lamp **10** forms the first light distribution area $A1$ having a predetermined light flux distribution using the first lens portion **61** for the far-side slit portion (the first slit portion **541** and the second slit portion **542**) and forms the second light distribution area $A2$ having a uniform light flux using the second lens portion **62** for the near-side slit portion (the third slit portion **543**). Therefore, even with the use of the single collecting lens **14**, the vehicle lamp **10** can form both a portion having a predetermined light flux distribution with modulation and a portion having a uniform light flux on the shade portion **51**. Therefore, the vehicle lamp **10** can make the middle of the leading end of the far-side radiation design (the first radiation design $Di1$ and the second radiation design $Di2$) clear as a center, can make the entire near-side radiation design (the third radiation design $Di3$) clear, and can form the more appropriate radiation pattern P .

[0095] The vehicle lamp **10** according to the first embodiment can obtain the following effects. The vehicle lamp **10** includes the first light source **31** and the second light source **32** arranged in parallel, the collecting lens **14** that collects light from them, the shade **15** as a shielding member

provided with the radiating slit **53** that partially passes the light collected thereby, and the projecting lens **16** that projects the light having passed therethrough to form the radiation pattern **Pi**. The collecting lens **14** includes the first lens portion **61** that corresponds to the first light source **31** and the second lens portion **62** that corresponds to the second light source **32**. The radiating slit **53** includes the near-side slit portion (the third slit portion **543**) that corresponds to the near-side radiation design (the third radiation design **Di3**) and the far-side slit portion (the first slit portion **541** and the second slit portion **542**) that corresponds to the far-side radiation pattern (the first radiation design **Di1** and the second radiation design **Di2**). The first lens portion **61** is provided to be opposed to the far-side slit portion, and the second lens portion **62** is provided to be opposed to the near-side slit portion. Therefore, the vehicle lamp **10** can form two light distribution areas (the first light distribution area **A1** and the second light distribution area **A2** according to the first embodiment) having different positions, sizes, and light (light flux) distributions by passing through two different optical paths of the first lens portion **61** and the second lens portion **62** using the single collecting lens **14** for the two light sources (**21**, **22**). Further, in the vehicle lamp **10**, the optical characteristics of the first lens portion **61** can be set in accordance with the light distribution area required on the far-side slit portion, and the optical characteristics of the second lens portion **62** can be set in accordance with the light distribution area required on the near-side slit portion. Therefore, the vehicle lamp **10** can efficiently use the light from the two light sources (**21**, **22**) and can irradiate the near-side slit portion and the far-side slit portion with a desired light flux distribution while simplifying the configurations of the first lens portion **61** and the second lens portion **62**. Therefore, the vehicle lamp **10** can easily adjust the light distribution on the shielding member and can form the radiation pattern **Pi** having a desired brightness distribution.

[0096] In the vehicle lamp **10**, the first lens portion **61** forms, on the shielding member (the shade **15**), the first light distribution area **A1** to irradiate the entire area of the far-side slit portion with the light from the first light source **31**, and the second lens portion **62** forms, on the shielding member (the shade **15**), the second light distribution area **A2** to irradiate the entire area of the near-side slit portion with the light from the second light source **32**. Therefore, in the vehicle lamp **10**, the positions and optical characteristics of the first light source **31** and the first lens portion **61** can be set to form the first light distribution area **A1**, and the positions and optical characteristics of the second light source **32** and the second lens portion **62** can be set to form the second light distribution area **A2**. Therefore, the vehicle lamp **10** can form the radiation pattern **Pi** having a desired brightness distribution while simplifying the configuration of the collecting lens **14**.

[0097] In the vehicle lamp **10**, the first lens portion **61** includes the curved incidence surface portion **65** opposed to the first light source **31**, the annular incidence surface portion **66** surrounding the curved incidence surface portion **65**, and the reflective surface **67** surrounding the annular incidence surface portion **66**. In addition, the second lens portion **62** is a convex lens that collects light from the second light source **32**. The first lens portion **61** forms the inner ring light distribution area **Ai** to irradiate the far-side slit portion with the light from the first light source **31** via the curved incidence surface portion **65** on the shielding member (the shade **15**). In addition, the first lens portion **61** forms, on the shielding member (the shade **15**), the outer ring light distribution area **Ao** to irradiate the entire area of the far-side slit portion while providing a light flux in the vicinity of the center of the far-side slit portion higher than surroundings with the light from the first light source **31** having passed through the annular incidence surface portion **66** and reflected by the reflective surface **67**. Therefore, the vehicle lamp **10** can form the outer ring light distribution area **Ao** and the inner ring light distribution area **Ai** having different positions, sizes, and light (light flux) distributions by using the difference in the optical path in the first lens portion **61**, can form the first light distribution area **A1** having a light flux distribution with a predetermined modulation by overlapping them, and can form a far-side radiation design having a desired brightness distribution.

[0098] In the vehicle lamp **10**, the second light distribution area **A2** has a lower light flux than the

inner ring light distribution area A_i and the outer ring light distribution area A_o and has a smaller light flux difference than the outer ring light distribution area A_o. For this reason, the vehicle lamp **10** can easily form the second light distribution area A₂ as described above by the second lens portion **62**, which is a convex lens, and can form a near-side radiation design having a desired brightness distribution.

[0099] In the vehicle lamp **10**, the first lens portion **61** and the second lens portion **62** are integrated. Therefore, in the vehicle lamp **10**, the relative positional accuracy between the first lens portion **61** and the second lens portion **62** can be improved, and the assembly process and the like can be facilitated.

[0100] With the vehicle lamp **10**, in the shielding member (the shade **15**), three quarters or more of the far-side slit portions (the first slit portion **541** and the second slit portion **542**) are provided below the projection optical axis L_p of the projecting lens **16**. For this reason, in the vehicle lamp **10**, the first lens portion **61** and the second lens portion **62** (the center positions (axis lines) thereof) can be provided to be vertically arranged to interpose the projection optical axis L_p in the collecting lens **14**, and the vehicle lamp **10** can have a small configuration as a whole.

[0101] In the vehicle lamp **10**, the first light source **31** and the second light source **32** are arranged in parallel in the vertical direction, and the first light source **31** is located below the second light source **32**. For this reason, the vehicle lamp **10** can be provided with the first light source **31** and the second light source **32** in accordance with the arrangement of the first lens portion **61** and the second lens portion **62** in the collecting lens **14**, and can have a small configuration as a whole.

[0102] In the vehicle lamp **10**, the far-side radiation design includes the first radiation design Di₁ and the second radiation design Di₂, and the far-side slit portion includes the first slit portion **541** that corresponds to the first radiation design Di₁ and the second slit portion **542** that corresponds to the second radiation design Di₂. In the vehicle lamp **10**, the first light distribution area A₁ provides, in the vicinity of the center of the first slit portion **541**, the highest light flux in the first light distribution area A₁ and the second light distribution area A₂. Therefore, the vehicle lamp **10** can irradiate the center of the far-side slit portion with the highest light flux, the center of the far-side radiation design of the radiation pattern P_i can be made clear, and the visibility can be improved by making the radiation pattern P_i have a desired brightness distribution.

[0103] Therefore, the vehicle lamp **10** according to the first embodiment as a vehicle lamp according to the present disclosure can form the radiation pattern P_i having a desired brightness distribution while efficiently using the light from both the light sources (**11**, **12**).

[0104] Although the vehicle lamp according to the present disclosure has been described above based on the first embodiment, the specific configuration is not limited to the first embodiment, and design changes, additions, and the like are allowed without departing from the gist of the invention according to each claim in the scope of claims.

[0105] According to the first embodiment, the radiation pattern P_i is configured such that the first radiation design Di₁ and the second radiation design Di₂, which are two V-shaped symbols, and the third radiation design Di₃, which has a corresponding pentagonal shape, are arranged at substantially equal intervals in the direction away from the vehicle **1**. However, as long as the radiation pattern is formed by the shade (shielding member) and includes a far-side radiation design and a near-side radiation design, the design of the symbol as the radiation design Di, the formation position, the number of the radiation designs Di, and the like, may be appropriately set, and are not limited to the configuration according to the first embodiment. The near-side radiation design may be projected onto a near position in the radiation pattern P_i, and the corresponding near-side slit portion may be provided to be opposed to the second lens portion **62**, and is not limited to the configuration according to the first embodiment. Further, the far-side radiation design may be projected onto a position farther than the near-side radiation design in the radiation pattern P_i, and the corresponding far-side slit portion may be provided to be opposed to the first lens portion **61**, and is not limited to the configuration according to the first embodiment.

[0106] Although the vehicle lamp **10** is provided in the front portion of the vehicle **1** according to the first embodiment, the vehicle lamp **10** may be accommodated in a door mirror, may be provided in a lamp chamber of a headlamp or a lamp chamber of a tail lamp (lamp chambers on both right and left sides of a rear portion of the vehicle), or may be provided in a vehicle body as long as it is provided in the vehicle **1** in accordance with the position where the radiation pattern is formed for the vehicle **1**, and is not limited to the configuration according to the first embodiment.

[0107] Furthermore, according to the first embodiment, the first light source **31** and the second light source **32** emit amber light. However, the color of the light emitted from the light source may be appropriately set in accordance with the arrangement location and the content to be conveyed, and is not limited to the configuration according to the first embodiment.

[0108] According to the first embodiment, the shade **15** that causes the light collected by the collecting lens **14** to pass through the radiating slit **53** is used as the shielding member. However, the shielding member may have a different configuration as long as the shielding member is provided with the radiating slit **53** that partially passes the light collected by the collecting lens **14**, and is not limited to the configuration according to each embodiment. As a different configuration, for example, it is possible to use a shielding plate (filter) in which a radiating slit that partially transmits light is provided in a plate-shaped film member that blocks the transmission of light and the light having passed through the collecting lens **14** is transmitted through the radiating slit.

[0109] According to the first embodiment, the vehicle lamp **10** is provided in the vehicle **1** driven by a driver. However, the vehicle lamp may be provided in a vehicle having an automatic driving function, and is not limited to the configuration according to the first embodiment. In this case, the vehicle lamp may form the radiation pattern at the timing corresponding to the usage for which the vehicle lamp is provided, i.e., at the timing corresponding to some intention regarding the operation of the vehicle **1**, and is not limited to the configuration according to the first embodiment.

[0110] In the configuration according to the first embodiment, the light source portion **12** is provided on the installation base portion **11** having a function as a heat sink, and the shielding frame **13**, the collecting lens **14**, the shade **15**, and the projecting lens **16** are attached to the installation base portion **11**. However, the vehicle lamp may have a different configuration as long as the light from the light source is collected on the shielding member by the collecting lens and projected by the projecting lens to form the radiation pattern, and is not limited to the configuration according to the first embodiment.

[0111] According to the first embodiment, the first light source **31** includes the two LED chips **31a** and the phosphor **31b** covering it, and the second light source **32** includes the one LED chip **32a** and the phosphor **32b** covering it. However, as long as the first light source **31** corresponds to the first lens portion **61** and the second light source **32** corresponds to the second lens portion **62**, the configuration of each of the two light sources may be appropriately set and is not limited to the configuration according to the first embodiment.

[0112] Hereinafter, a second embodiment of a vehicle lamp **110** as an example of a vehicle lamp according to the present disclosure will be described with reference to the drawings. In FIG. **13**, the vehicle lamp **110** is emphasized with respect to a vehicle **101** in order to facilitate understanding of the state in which the vehicle lamp **110** is provided, and does not necessarily coincide with the actual state. Further, in FIG. **21**, only the periphery of a shade portion **133** (each slit portion **136** thereof) is illustrated in order to facilitate understanding of a state in which the second light distribution area is formed in a second slit portion **1362** by the light from a second light source **125** on a shade **114** (a radiating slit **135** of the shade portion **133**) and an unintended light distribution area (stray light area) is formed in a third slit portion **1363**. In each area illustrated in FIG. **21**, the area corresponding to the level of the light flux (light amount) is surrounded by a broken line, and the light (light flux) distribution is illustrated like a contour line in which the light flux increases toward the center of the area.

Second Embodiment

[0113] The vehicle lamp **110** according to the second embodiment, which is an embodiment of the vehicle lamp according to the present disclosure, will be described with reference to FIGS. **13** to **21**. As illustrated in FIG. **13**, the vehicle lamp **110** according to the second embodiment is used as a lamp of the vehicle **101** such as an automobile and is provided in a front portion of the vehicle **101** to form the radiation pattern P_i on a road surface **102** in the vicinity of the front of the vehicle **101**, separately from a headlamp provided in the vehicle **101**. The vicinity of the front of the vehicle **101** always includes a proximity area closer to the vehicle **101** than a headlight area irradiated by the headlight provided in the vehicle **101** and may partially include the headlight area. The vehicle lamp **110** may also form the radiation pattern P_i on the road surface **102** in the vicinity of the rear or side of the vehicle **101** and is not limited to the configuration according to the second embodiment.

[0114] According to the second embodiment, each of the vehicle lamps **110** is provided on the front end of the vehicle **101** at a position higher than the road surface **102** and is provided in a state where the projection optical axis L_p (see FIG. **14** and the like) is inclined with respect to the road surface **102**. The two vehicle lamps **110** have basically the same configuration except that the mounting positions and the positions where the radiation patterns P_i are formed are different. In the description below, in each of the vehicle lamps **110**, the direction in which the projection optical axis L_p extends, i.e., the direction in which light is irradiated (projected), is an optical axis direction (Z in the drawings), a vertical direction when the optical axis direction extends along a horizontal plane is an up-down direction (Y in the drawings), and the direction (horizontal direction) orthogonal to the optical axis direction and the up-down direction is a width direction (X in the drawings) (see FIG. **14** and the like).

[0115] As illustrated in FIG. **14**, in the vehicle lamp **110**, a light source portion **112**, a collecting lens **113**, the shade **114**, and a projecting lens **115** are attached to an installation base portion **111** to form a single projection optical system and constitute a road surface projection unit of a projector type. The installation base portion **111** is a portion where the light source portion **112** is provided, is made of aluminum die-casting or resin having thermal conductivity, and functions as a heat sink that releases the heat generated in the light source portion **112** to the outside as a whole. The installation base portion **111** includes a base portion **121** and a pair of mounting arm portions **122**.

[0116] The base portion **121** has a flat plate shape orthogonal to the optical axis direction and, the light source portion **112** is mounted on a light source mounting area at the center. The light source mounting area has a flat surface and is provided with a plurality of screw holes and a plurality of positioning protrusions. In addition, the base portion **121** is provided with a plurality of heat radiation fins, and the heat generated in the light source portion **112** installed at the light source mounting area is radiated to the outside mainly from each of the heat radiation fins.

[0117] The pair of mounting arm portions **122** is provided as a pair on both outer sides in the width direction of the light source portion **112** and protrudes forward from the base portion **121** in the optical axis direction. End portions of the two mounting arm portions **122** on the front side in the optical axis direction are flat surfaces orthogonal to the optical axis direction. Each of the end portions is provided with a positioning protrusion **122a** and a screw hole **122b**. The positioning protrusion **122a** is provided in a lower portion of the end portion in the up-down direction and protrudes forward in the optical axis direction. The screw hole **122b** is provided in an upper portion of the end portion in the up-down direction, and the collecting lens **113**, the shade **114**, and the projecting lens **115** can be fixed by screwing a screw **123**.

[0118] The light source portion **112** includes a first light source **124**, a second light source **125**, a third light source **126** (see FIG. **15** and the like), a connector terminal **127**, and a substrate **128** on which these components are mounted. The three light sources (**124**, **125**, **126**) include a light emitting element such as a light emitting diode (LED). According to the second embodiment, the three light sources (**124**, **125**, **126**) emit light in amber color (amber light) with a Lambertian distribution centered on the emission optical axis. The colors (wavelength bands), the mode of

distribution, the number of colors, and the like, of the three light sources (**124**, **125**, **126**) may be appropriately set and are not limited to the configuration according to the second embodiment. [0119] As illustrated in FIG. **15** and the like, the three light sources (**124**, **125**, **126**) according to the second embodiment are arranged in parallel in the up-down direction with the projection optical axis **Lp** interposed therebetween, the first light source **124** is located below the projection optical axis **Lp**, the second light source **125** is located above the projection optical axis **Lp**, and the third light source **126** is located above the second light source **125**. The first light source **124** has a rectangular shape elongated in the width direction and includes two LED chips **124a** and a phosphor **124b** covering each of them. The second light source **125** has a substantially square shape and includes one LED chip **125a** and a phosphor **125b** covering it. The third light source **126** has a substantially square shape and includes one LED chip **126a** and a phosphor **126b** covering it. The three light sources (**124**, **125**, **126**) emit light from the LED chips **124a**, **125a**, **126a** as amber light by passing the light through the phosphors **124b**, **125b**, **126b**. Therefore, in the first light source **124**, the phosphor **124b** functions as a first light emission surface, in the second light source **125**, the phosphor **125b** functions as a second light emission surface, and in the third light source **126**, the phosphor **126b** functions as a third light emission surface.

[0120] As illustrated in FIG. **14**, the connector terminal **127** is electrically connected to a wiring pattern of the substrate **128**, and a connection connector connected to a lighting control circuit is detachably attached. The connector terminal **127** is provided at a lower end portion of the substrate **128** in the up-down direction so that the connection connector can be easily attached and detached. When the connection connector is attached to the connector terminal **127**, electric power can be supplied from the lighting control circuit to the light sources (**124**, **125**, **126**) via the wiring pattern.

[0121] The substrate **128** has a plate shape formed of a resin material such as a glass epoxy substrate, and each of the light sources (**124**, **125**, **126**) is mounted thereon. In the substrate **128**, a plurality of screw through-holes is provided corresponding to the respective screw holes of the light source mounting area of the base portion **121** of the installation base portion **111**, and a plurality of positioning holes is provided corresponding to the respective positioning protrusions of the light source mounting area. The substrate **128** is attached to the base portion **121** by inserting the corresponding positioning protrusions to the positioning holes and screwing screws **129** inserted into the screw through-holes into the corresponding screw holes. Accordingly, the substrate **128** causes the mounted light sources (**124**, **125**, **126**) to be opposed to the collecting lens **113**. The substrate **128** appropriately supplies power from the lighting control circuit via the connector terminal **127** to appropriately turn on the light sources (**124**, **125**, **126**).

[0122] The collecting lens **113** collects the light emitted from each of the light sources (**124**, **125**, **126**) and collects the light in the periphery of each of the slit portions **136** described below on the shade **114**, i.e., in the area where each of the slit portions **136** is provided while including all the slit portions **136** on the shade **114**. The collecting lens **113** includes a collecting lens main body **131** that collects light from each of the light sources (**124**, **125**, **126**) and a pair of collecting lens mounting pieces **132** that protrudes from the collecting lens main body in the width direction. The collecting lens main body **131** and the collecting lens mounting piece **132** are integrally formed, and are integrally formed by resin molding using a mold according to the second embodiment. Optical characteristics of the collecting lens main body **131** are set to form a predetermined light distribution area on the shade **114**. This will be described below.

[0123] The two collecting lens mounting pieces **132** are formed in a plate shape orthogonal to the optical axis direction and can be applied to the end portions of the two mounting arm portions **122** of the base portion **121** of the installation base portion **111**. Each of the collecting lens mounting pieces **132** is provided with a collecting lens positioning hole **132a** and a collecting lens screw through-hole **132b**. The positioning protrusion **122a** can be fitted into each of the collecting lens positioning holes **132a** in a state where the collecting lens mounting piece **132** is applied to the end portion. The screws **123** screwed into the screw holes **122b** can be inserted into the respective

collecting lens screw through-holes **132b** in a state where the collecting lens mounting piece **132** is applied to the end portion. The collecting lens **113** is attached to the two mounting arm portions **122** (the end portions thereof) of the installation base portion **111** by inserting the corresponding positioning protrusions **122a** into the collecting lens positioning holes **132a** and screwing the screws **123** inserted into the collecting lens screw through-holes **132b** into the corresponding screw holes **122b**.

[0124] The shade **114** is an example of a shielding member that forms the radiation pattern P_i by causing the light from each of the light sources (**124**, **125**, **126**) collected by the collecting lens **113** to partially pass through the radiating slit **135** described below. As illustrated in FIG. **13**, in the radiation pattern P_i , the three radiation designs D_i are arranged at substantially equal intervals in a direction away from the vehicle **101**. In this case, when each of the radiation designs D_i is individually illustrated, the radiation design farthest from the vehicle **101** is defined as the first radiation design D_{i1} , and the radiation designs are defined as the second radiation design D_{i2} and the third radiation design D_{i3} in order as they are closer to the vehicle **101**. According to the second embodiment, each of the radiation designs D_i has a substantially isosceles triangle shape having a base on the vehicle **101** side and has substantially the same size and the same shape.

[0125] The radiation pattern P_i is formed by arranging the first radiation design D_{i1} , the second radiation design D_{i2} , and the third radiation design D_{i3} on the same straight line so as to be away from the vehicle **101** on the road surface **102** as a projection surface. Thus, the radiation pattern P_i can look like an arrow indicating the arranged direction by arranging the three radiation designs D_i . The direction indicated by the arrow as the radiation pattern P_i , i.e., the direction in which the apexes of the substantially isosceles triangles of the radiation designs D_i are arranged is the arrow direction D_a , and the indicated side (the first radiation design D_{i1} side) is the front side of the arrow direction D_a . The radiation pattern P_i including the three radiation designs D_i is formed by the shade **114**.

[0126] As illustrated in FIG. **14**, the shade **114** is made of a plate-like member that basically blocks transmission of light and includes a shade portion **133** and a pair of shade mounting pieces **134**. The shade mounting pieces **134** protrude from both sides of the shade portion **133** in the width direction and can be applied to the respective collecting lens mounting pieces **132** of the collecting lens **113** attached to the end portions of the two mounting arm portions **122** of the installation base portion **111**. Each of the shade mounting pieces **134** is provided with a shade positioning hole **134a** and a shade screw through-hole **134b**. Each of the shade positioning holes **134a** allows the positioning protrusion **122a** passed therethrough to be fitted into the shade positioning hole **134a** in a state where the shade mounting piece **134** is applied to the collecting lens mounting piece **132**. Each of the shade screw through-holes **134b** allows the screw **123** inserted into the collecting lens screw through-hole **132b** to be inserted into the shade screw through-hole **134b** in a state where the shade mounting piece **134** is applied to the collecting lens mounting piece **132**. The shade **114** is attached to the two mounting arm portions **122** of the installation base portion **111** via the collecting lens **113** by inserting the corresponding positioning protrusion **122a** to each of the shade positioning holes **134a** and screwing each of the screws **123** inserted into each of the shade screw through-holes **134b** into the corresponding screw hole **122b**. In the shade **114**, the center position of the shade portion **133** is located on the projection optical axis L_p by attaching the shade mounting piece **134** to the two mounting arm portions **122**.

[0127] The shade portion **133** is provided with the radiating slit **135** formed by partially cutting out and penetrating the plate-like member. The radiating slit **135** partially transmits the light from each of the light sources (**124**, **125**, **126**) collected by the collecting lens **113** (the collecting lens main body **131** thereof), thereby forming the radiation pattern P_i to be projected into a predetermined shape. The radiating slit **135** corresponds to the radiation pattern P_i , and includes the three slit portions **136** according to the second embodiment.

[0128] The three slit portions **136** correspond one to one to the three radiation designs D_i . Since the

projecting lens **115** inverts the shade **114** (the radiating slit **135**) and projects the inverted shade **114** onto the road surface **102**, each of the slit portions **136** has a rotationally symmetric positional relationship around the projection optical axis L_p with respect to the positional relationship of each of the radiation designs D_i of the radiation pattern P_i . Therefore, in each of the slit portions **136**, a first slit portion **1361** on the lowermost side in the up-down direction corresponds to the first radiation design D_{i1} of the radiation pattern P_i , a second slit portion **1362** above it corresponds to the second radiation design D_{i2} , and a third slit portion **1363** above it corresponds to the third radiation design D_{i3} .

[0129] The position and the size of each of the slit portions **136** on the shade portion **133** are set such that each of the radiation designs D_i has a target size and a target positional relationship on the road surface **102**. In the shade **114** according to the second embodiment, in the up-down direction, the third slit portion **1363** is provided above the projection optical axis L_p , the second slit portion **1362** is provided on the projection optical axis L_p , and the first slit portion **1361** is provided below the second slit portion (see FIG. 21). The light transmitted through the shade **114** (each of the slit portions **136** of the radiating slit **135**) is projected onto the road surface **102** by the projecting lens **115**.

[0130] The slit portions **136** are formed in a substantially isosceles triangle shape similarly to the respective corresponding radiation designs D_i and are vertically and horizontally inverted with respect to the respective radiation designs D_i . The sizes and the intervals of the three slit portions **136** are set in accordance with the distances to the road surface **102** so that the radiation designs D_i have the sizes illustrated in FIG. 13 and are arranged at substantially equal intervals on the road surface **102**. Specifically, in the vehicle lamp **110**, since the projection optical axis L_p is provided to be inclined with respect to the road surface **102**, the distances from the shade **114** and the projecting lens **115** to the road surface **102** are different, and therefore, when projected onto the road surface **102** by the projecting lens **115**, each of the slit portions **136** (each of the radiation designs D_i that is light transmitted therethrough) has the size and the interval corresponding to the distance. Specifically, according to the second embodiment, the first slit portion **1361** is a substantially isosceles triangle that is the smallest and is contracted in the height direction, the second slit portion **1362** is a substantially isosceles triangle that is larger than the first slit portion **1361**, and the third slit portion **1363** is a substantially isosceles triangle that is larger than the second slit portion **1362** and is curved so that the bottom surface slightly swells. Each of the slit portions **136** is wider than the corresponding radiation design D_i in the width direction, and the interval between the second slit portion **1362** and the third slit portion **1363** is larger than the interval between the first slit portion **1361** and the second slit portion **1362**.

[0131] In this way, the three slit portions **136** are different in size from the respective radiation designs D_i and are arranged at different intervals. In each of the slit portions **136**, the first slit portion **1361** has the smallest reduction ratio with respect to the corresponding radiation design D_i , and the passed light is enlarged at the largest enlargement ratio when projected on the road surface **102** to form the first radiation design D_{i1} . In addition, in each of the slit portions **136**, the third slit portion **1363** has the largest reduction ratio with respect to the corresponding radiation design D_i , and the passed light is enlarged at the smallest enlargement ratio when being projected onto the road surface **102** to form the third radiation design D_{i3} .

[0132] The projecting lens **115** includes a projecting lens main body **137** that projects light having passed through the shade **114** and a pair of projecting lens mounting pieces **138** that protrudes from it in the width direction. The projecting lens main body **137** is a convex lens having a circular shape when viewed in the optical axis direction and, according to the second embodiment, is a free-form surface in which the incidence surface and the emission surface are convex surfaces. The projecting lens main body **137** forms the radiation pattern P_i on the road surface **102** inclined with respect to the projection optical axis L_p by projecting the radiating slit **135** (each of the slit portions **136** thereof) of the shade **114** (see FIG. 13). The incidence surface and the emission surface may be

convex or concave surfaces as long as the projecting lens main body **137** is a convex lens and are not limited to the configuration according to the second embodiment.

[0133] The two projecting lens mounting pieces **138** are formed in a plate shape orthogonal to the optical axis direction and can be applied to the shade mounting pieces **134** of the shade **114** attached to the end portions of the two mounting arm portions **122** of the installation base portion **111**. Each of the projecting lens mounting pieces **138** is provided with a projecting lens positioning hole **138a** and a projecting lens screw through-hole. Each of the projecting lens positioning holes **138a** allows the positioning protrusion **122a** passed therethrough to be fitted into the projecting lens positioning hole **138a** in a state where the projecting lens mounting piece **138** is applied to the shade mounting piece **134**. Each of the projecting lens screw through-holes allows the screw **123** inserted through the shade screw through-hole **134b** to be inserted into the projecting lens screw through-hole in a state where the projecting lens mounting piece **138** is applied to the shade mounting piece **134**. The projecting lens **115** is attached to the two mounting arm portions **122** (the end portions thereof) of the installation base portion **111** by inserting the corresponding positioning protrusion **122a** into each of the projecting lens positioning holes **138a** and screwing each of the screws **123** inserted into the respective projecting lens screw through-holes into the corresponding screw hole **122b**. Accordingly, in the projecting lens **115**, the projection optical axis L_p , which is the optical axis of the projecting lens main body **137**, is directed in a predetermined direction, and the direction of the projection optical axis L_p of the vehicle lamp **110** is set.

[0134] Next, the configuration of the collecting lens main body **131** of the collecting lens **113** will be described mainly with reference to FIGS. **16** to **19**. The collecting lens main body **131** (the collecting lens **113**) includes a front surface **141** opposed to the shade **114** and a back surface **142** opposed to the light source portion **112**. The collecting lens main body **131** includes a first lens portion **143** that corresponds to the first light source **124**, a second lens portion **144** that corresponds to the second light source **125**, and a third lens portion **145** that corresponds to the third light source **126**. In the collecting lens **113** (the collecting lens main body **131**) according to the second embodiment, the first lens portion **143**, the second lens portion **144**, and the third lens portion **145** are integrally formed in a state where the second lens portion **144** is placed on the first lens portion **143** and the third lens portion **145** is placed on the second lens portion **144**. In the collecting lens **113** according to the second embodiment, the (optical) shapes of the front surface **141** side and the back surface **142** side of the first lens portion **143**, the second lens portion **144**, and the third lens portion **145** are set so that each of the slit portions **136** of the shade **114** is appropriately irradiated with the light emitted from the light source portion **112**, i.e., each light source (**124**, **125**, **126**).

[0135] The first lens portion **143** is opposed to the first light source **124** in the optical axis direction (located on the emission optical axis of the first light source **124**) and collects the light from the first light source **124** to an area where the first slit portion **1361** of the shade **114** is provided. As illustrated in FIGS. **16**, **17**, **19** and the like, the back surface **142** of the first lens portion **143** has a central portion recessed toward the inside of the collecting lens **113** (the side opposite to the light source portion **112**) and is provided with a first opposing incidence surface portion **151** curved convexly outward at the center, a first inclined incidence surface portion **152** surrounding it, and a first reflective surface **153** surrounding the first inclined incidence surface portion **152** in a truncated cone shape.

[0136] The first opposing incidence surface portion **151** is provided on the emission optical axis to be opposed to the first light source **124** in the optical axis direction, and the first light source **124** is located in the vicinity of the focal point on the rear side (rear focal point). The first opposing incidence surface portion **151** causes the light emitted from the first light source **124** to enter the first lens portion **143** as parallel light traveling substantially parallel to the axis line of the first lens portion **143**, and causes the light to travel toward a first inner emission surface portion **155** (described below) of the front surface **141**. In addition, the parallel light (light that is parallel) refers

to the light that is collimated by passing through the first opposing incidence surface portion **151**. [0137] The first inclined incidence surface portion **152** is provided to protrude toward the first light source **124** side and causes the light which does not travel to the first opposing incidence surface portion **151** among the light from the first light source **124** to be incident on the first lens portion **143**. The first reflective surface **153** is provided at a position where the light incident on the collecting lens **113** from the first inclined incidence surface portion **152** travels. The first reflective surface **153** reflects light incident from the first inclined incidence surface portion **152** and causes the light to travel toward a first outer emission surface portion **156** described below of the front surface **141** as parallel light traveling substantially parallel to the axis line of first lens portion **143**. The first reflective surface **153** may reflect light by using total reflection, or may reflect light by adhering aluminum, silver, or the like by vapor deposition, coating, or the like. For these reasons, the back surface **142** causes the light emitted from the first light source **124** to be efficiently incident and guides the light to the front surface **141**.

[0138] Therefore, in the first lens portion **143**, the light having passed through the first opposing incidence surface portion **151** becomes direct light that travels directly toward the front surface **141** side, and the light having passed through the first inclined incidence surface portion **152** and reflected by the first reflective surface **153** becomes reflected light that travels toward the front surface **141** side after being internally reflected. Since the first lens portion **143** has such a configuration, it is possible to efficiently use light emitted from the corresponding first light source **124**.

[0139] The first lens portion **143** is provided with a first emission surface **154** that emits light incident on the front surface **141** side to the front side in the front-back direction. As illustrated in FIG. **18** and the like, the first emission surface **154** has a circular shape in which a part of the upper side is cut out in a front view, and includes the first inner emission surface portion **155** and the first outer emission surface portion **156** having different optical settings. The first inner emission surface portion **155** is provided in an area of the first emission surface **154** where the light having passed through the first opposing incidence surface portion **151** travels, and has a substantially circular shape in a front view. The first inner emission surface portion **155** protrudes outward (the projecting lens **115** side (the front side in the front-back direction)) of the collecting lens **113** beyond the first outer emission surface portion **156**. The first inner emission surface portion **155** refracts the light emitted from the first light source **124** and passing through the first opposing incidence surface portion **151**, thereby forming a plurality of light distribution images of the first light source **124** on the first slit portion **1361** of the shade **114** at positions corresponding to the optical characteristics in an appropriately overlapping manner. The optical characteristics can be set by adjusting the curvature (surface shape) of the first inner emission surface portion **155** together with the first opposing incidence surface portion **151** for each place and, according to the second embodiment, the curvatures are set to be gradually changed.

[0140] The first outer emission surface portion **156** is provided to surround the area sandwiching the first inner emission surface portion **155** in the width direction and the area below the first inner emission surface portion **155**, and is located in the area where the light travels after being emitted from the first light source **124**, passing through the first inclined incidence surface portion **152**, and being reflected by the first reflective surface **153**. The first outer emission surface portion **156** is located (recessed) on the inner side (the back side in the front-back direction) of the collecting lens **113** with respect to the first inner emission surface portion **155**. The first outer emission surface portion **156** refracts the light emitted from the first light source **124**, passed through the first inclined incidence surface portion **152**, and reflected by the first reflective surface **153**, thereby forming a plurality of light distribution images of the first light source **124** on the first slit portion **1361** of the shade **114** at positions corresponding to the optical characteristics in an appropriately overlapping manner. The optical characteristics can be set by adjusting the curvature (surface shape) of the first outer emission surface portion **156** together with the first reflective surface **153**

for each place, and according to the second embodiment, the curvatures are set to be gradually changed.

[0141] For example, the first lens portion **143** forms a light distribution area to brighten the entire area of the first slit portion **1361** with the light having passed through the first inner emission surface portion **155** and forms a light distribution area to brighten the vicinity of the apex of the first slit portion **1361** with the light having passed through the first outer emission surface portion **156**. Then, the first lens portion **143** overlaps the light having passed through the first inner emission surface portion **155** with the light having passed through the first outer emission surface portion **156** and thus forms a first light distribution area in which the vicinity of the apex of the first slit portion **1361** has the highest light flux (light amount) while the entire area of the first slit portion **1361** is brightened. As long as the first lens portion **143** forms the light distribution area that brightens the first slit portion **1361** from the viewpoint of appropriately forming the corresponding first radiation design Di1, the brightness distribution, shape, and the like in the light distribution areas formed by the first inner emission surface portion **155** and the first outer emission surface portion **156** may be appropriately set, and are not limited to the second embodiment.

[0142] The second lens portion **144** is opposed to the second light source **125** in the optical axis direction (located on the emission optical axis of the second light source **125**), and collects the light from the second light source **125** to the area where the second slit portion **1362** of the shade **114** is provided. The second lens portion **144** has the same basic configuration on the back surface **142** and the front surface **141** and the same basic concept of light collection with respect to the second slit portion **1362** except for detailed optical settings caused by the difference in size and shape of the corresponding slit portion **136**. Therefore, hereinafter, the configuration and optical setting of the second lens portion **144** will be briefly described.

[0143] As illustrated in FIGS. **16**, **17**, **19** and the like, the back surface **142** of the second lens portion **144** has a central portion recessed toward the inside of the collecting lens **113** (the side opposite to the light source portion **112**) and is provided with a second opposing incidence surface portion **157** curved convexly outward at the center, a second inclined incidence surface portion **158** surrounding it, and a second reflective surface **159** surrounding the second inclined incidence surface portion **158** in a truncated cone shape. The second opposing incidence surface portion **157** is provided on the emission optical axis to be opposed to the second light source **125** in the optical axis direction, causes the light emitted from the second light source **125** to be incident on the collecting lens **113** as parallel light traveling substantially parallel to the axis line of the second lens portion **144**, and causes the light to travel toward a second inner emission surface portion **162** described below of the front surface **141**. The second inclined incidence surface portion **158** is provided to protrude toward the second light source **125** side, and causes the light that does not travel to the second opposing incidence surface portion **157** among the light from the second light source **125** to be incident on the second lens portion **144**. The second reflective surface **159** reflects the light incident on the collecting lens **113** from the second inclined incidence surface portion **158**, and collects the light so that the light travels toward a second outer emission surface portion **163** described below of the front surface **141** as parallel light traveling substantially parallel to the axis line of the second lens portion **144**.

[0144] The second lens portion **144** is provided with a second emission surface **161** that emits light incident on the front surface **141** side to the front side in the front-back direction. The second emission surface **161** has a substantially rectangular shape elongated in the width direction in a front view, and includes a second inner emission surface portion **162** and a second outer emission surface portion **163** having different optical settings. The second inner emission surface portion **162** is provided in an area of the second emission surface **161** where the light having passed through the second opposing incidence surface portion **157** travels, has a substantially circular shape in a front view, and protrudes outward (the projecting lens **115** side (the front side in the front-back direction)) of the collecting lens **113** beyond the second outer emission surface portion **163**. The

second outer emission surface portion **163** is provided in the area sandwiching the second inner emission surface portion **162** in the width direction, is located in the area where the light travels after being emitted from the second light source **125**, passing through the second inclined incidence surface portion **158**, and being reflected by the second reflective surface **159**, and is located (recessed) on the inner side (the back side in the front-back direction) of the collecting lens **113** with respect to the second inner emission surface portion **162**.

[0145] For example, the second lens portion **144** forms a light distribution area to brighten the entire area of the second slit portion **1362** with the light having passed through the second inner emission surface portion **162** and forms a light distribution area to brighten the vicinity of the apex of the second slit portion **1362** with the light having passed through the second outer emission surface portion **163**. Then, the second lens portion **144** overlaps the light having passed through the second inner emission surface portion **162** with the light having passed through the second outer emission surface portion **163** and thus forms a second light distribution area AL (see FIG. **21**) in which the vicinity of the apex of the second slit portion **1362** has the highest light flux (light amount) while the entire area of the second slit portion **1362** is brightened. As long as the second lens portion **144** forms the light distribution area that brightens the second slit portion **1362** from the viewpoint of appropriately forming the corresponding first radiation design Di1, the brightness distribution, shape, and the like in the light distribution area formed by the second inner emission surface portion **162** and the second outer emission surface portion **163** may be appropriately set, and are not limited to the second embodiment.

[0146] As illustrated in FIGS. **16** to **19** and the like, the third lens portion **145** is a convex lens formed in a substantially rectangular shape elongated in the width direction when viewed from the front in the optical axis direction and, as a whole, collects the light emitted from the third light source **126** and having a spread in a state close to being parallel to the projection optical axis Lp and causes the light to travel to the shade portion **133**. The third lens portion **145** includes a third incidence surface **164** opposed to the third light source **126** and a third emission surface **165** directed to the opposite side. According to the second embodiment, the third lens portion **145** is a free-form surface in which the third incidence surface **164** and the third emission surface **165** are convex surfaces. The third incidence surface **164** and the third emission surface **165** may be a convex surface or a concave surface as long as the third lens portion **145** is a convex lens, and are not limited to the configuration according to the second embodiment.

[0147] The third incidence surface **164** is opposed to the third light source **126** in the optical axis direction, and the third light source **126** is located in the vicinity of the focal point on the rear side (rear focal point). The third incidence surface **164** causes the light emitted from the third light source **126** to enter the third lens portion **145** as parallel light traveling substantially parallel to the axis line of the third lens portion **145**. The third emission surface **165** is provided on the side opposite to the third incidence surface **164**, and refracts the light having passed through the third incidence surface **164** to cause the light to travel toward the front side in the front-back direction while diffusing the light. The third emission surface **165** emits the light from the third light source **126** through the third incidence surface **164** to form a plurality of light distribution images of the third light source **126** at positions corresponding to the optical characteristics on the shade **114** (the shade portion **133**) in an appropriately overlapping manner. The optical characteristics can be set by adjusting the curvatures (surface shapes) of the third incidence surface **164** and the third emission surface **165** for each place and, according to the second embodiment, the curvatures are set to be gradually changed.

[0148] The third emission surface **165** appropriately refracts the light emitted from the third light source **126** and passing through the third incidence surface **164**, thereby forming a third light distribution area in which the entire area of the third slit portion **1363** of the shade **114** has a substantially equal light flux (light amount). Here, a substantially equal light flux in the entire area means that a change in the light flux is smaller than that in at least the first light distribution area or

the second light distribution area AL, and preferably means a substantially uniform light flux. It is assumed that the third light distribution area according to the second embodiment has a lower light flux than the first light distribution area and the second light distribution area AL.

[0149] In the collecting lens **113**, the first lens portion **143** forms the first light distribution area in the first slit portion **1361** with the light from the first light source **124**, the second lens portion **144** forms the second light distribution area AL in the second slit portion **1362** with the light from the second light source **125**, and the third lens portion **145** forms the third light distribution area in the third slit portion **1363** with the light from the third light source **126**. Therefore, on the shade **114**, the collecting lens **113** can form a predetermined light flux distribution with a predetermined modulation (inflection (differences in the level of the light flux)) for the first slit portion **1361** and the second slit portion **1362** and can form a uniform light flux distribution lower than that of the first slit portion **1361** and the second slit portion **1362** for the third slit portion **1363**.

[0150] In the collecting lens **113**, the collecting lens main body **131** is provided with a first light reduction portion **171** and a second light reduction portion **172**. The first light reduction portion **171** and the second light reduction portion **172** reduce or block light (light flux) from each light source (**124**, **125**, **126**) in the collecting lens main body **131**, and the first light reduction portion **171** is located between the first lens portion **143** and the second lens portion **144**, and the second light reduction portion **172** is located between the second lens portion **144** and the third lens portion **145**.

[0151] According to the second embodiment, the first light reduction portion **171** includes a front-surface-side first light reduction portion **173** and a back-surface-side first light reduction portion **174**, and the second light reduction portion **172** includes a front-surface-side second light reduction portion **175** and a back-surface-side second light reduction portion **176**. The front-surface-side first light reduction portion **173** is formed, on the front surface **141** of the collecting lens main body **131**, by performing light reduction processing between the first emission surface **154** of the first lens portion **143** and the second emission surface **161** of the second lens portion **144**. In addition, the front-surface-side second light reduction portion **175** is formed, on the front surface **141** of the collecting lens main body **131**, by performing light reduction processing between the second emission surface **161** of the second lens portion **144** and the third emission surface **165** of the third lens portion **145**. Further, the back-surface-side first light reduction portion **174** is formed, on the back surface **142** of the collecting lens main body **131**, by performing light reduction processing between the first opposing incidence surface portion **151**, the first inclined incidence surface portion **152**, and the first reflective surface **153** of the first lens portion **143** and the second opposing incidence surface portion **157**, the second inclined incidence surface portion **158**, and the second reflective surface **159** of the second lens portion **144**. The back-surface-side second light reduction portion **176** is formed, on the back surface **142** of the collecting lens main body **131**, by performing light reduction processing between the second opposing incidence surface portion **157**, the second inclined incidence surface portion **158**, and the second reflective surface **159** of the second lens portion **144** and the third incidence surface **164** of the third lens portion **145**.

[0152] Examples of the light reduction processing include forming a shielding surface by applying a light-shielding coating material such as black, and forming a scattering surface by forming a fine optical element, a prism (unevenness), emboss, or the like. In the collecting lens main body **131**, the first light reduction portion **171** and the second light reduction portion **172** are provided by performing the above-described light reduction processing on a band-shaped area having a predetermined size in the up-down direction while extending in the width direction with respect to the above-described position on the front surface **141** or the back surface **142**. Here, the size of the first light reduction portion **171** and the second light reduction portion **172** in the up-down direction is equal to or less than 1 mm, and is 0.5 mm according to the second embodiment.

[0153] In addition, according to the second embodiment, in the third lens portion **145**, the light reduction processing is also performed on a bottom surface **145a** located on the second lens portion

144 side, and a bottom-surface-side second light reduction portion **177** forming a part of the second light reduction portion **172** is provided. While the third lens portion **145** is a convex lens and the third incidence surface **164** is a free-form surface as a convex surface as described above, the second lens portion **144** includes the second opposing incidence surface portion **157**, the second inclined incidence surface portion **158**, and the second reflective surface **159**, and thus the bottom surface **145a** is formed by the difference in shape therebetween (see FIG. **18**). Since the bottom surface **145a** is located obliquely forward and upward of the second light source **125**, the light directly traveling from the second light source **125** may enter the third lens portion **145**. Therefore, according to the second embodiment, the bottom-surface-side second light reduction portion **177** is formed by also performing the light reduction processing on the bottom surface **145a** of the third lens portion **145**.

[0154] Next, the operation of the vehicle lamp **110** will be described. The vehicle lamp **110** can turn on and off the light sources (**124**, **125**, **126**) individually, sequentially, or simultaneously by supplying electric power from the lighting control circuit to the light sources from the substrate **128**. The light from each of the light sources (**124**, **125**, **126**) is collected by the collecting lens **113**, is emitted to the shade **114**, passes through the radiating slit **135** (each of the slit portions **136**), and is then projected by the projecting lens **115**, thereby forming the radiation pattern P_i on the road surface **102**. The light transmitted through the radiating slit **135** (each of the slit portions **136** thereof) of the shade **114** having the above-described light (light flux) distribution is projected by the projecting lens **115**, and thus the radiation pattern P_i is formed individually, sequentially, or simultaneously at the positions where the three radiation designs D_i are arranged in the arrow direction D_a .

[0155] The vehicle lamp **110** is interlocked with the turn lamps, and when one of the left and right turn lamps is turned on, the light sources (**124**, **125**, **126**) provided on the turned-on side are turned on to form the radiation pattern P_i on the road surface **102**. According to the second embodiment, first, the third light source **126** is turned on, then the second light source **125** is turned on while the third light source **126** is continuously turned on, then the first light source **124** is turned on while the third light source **126** and the second light source **125** are continuously turned on, then the light sources (**124**, **125**, **126**) are turned off all at once, and thereafter this is repeated. As a result, the radiation pattern P_i can appear as if the radiation pattern P_i extends in the arrow direction D_a by turning on the third radiation design D_{i3} , the second radiation design D_{i2} , and the first radiation design D_{i1} in this order. Therefore, in a scene where the vehicle **101** is traveling from an alley with poor visibility to a different alley, the vehicle lamp **110** can cause a person on the different alley to visually recognize the radiation pattern P_i formed on the road surface **102** even when the person cannot visually recognize the vehicle **101**. In addition, in the vehicle **101**, when the hazard lamps are turned on, the two left and right vehicle lamps **110** simultaneously form the radiation patterns P_i on the road surface **102** as described above and, therefore, it is possible to more reliably recognize that the hazard lamps are turned on, compared to a case where only the left and right turn lamps are blinked.

[0156] Here, in the conventional vehicle lamp described in the related art document, a plurality of light guides is provided individually corresponding to a plurality of light sources, and light from each light source is efficiently used. In the conventional vehicle lamp, each of the light guide members diffuses light therein to emit light having a substantially uniform light flux distribution, and a shade (shielding member) is irradiated with the light passing through each of the light guide members to obtain a substantially uniform light (light flux) distribution on the shade. Further, in the conventional vehicle lamp, since the light from the light source corresponding to each light guide is guided onto the shade, the light is separately guided onto the shade for each light source. For this reason, it is difficult for the conventional vehicle lamp to obtain a desired light distribution as in the case where an area having a high light flux is partially formed while continuously changing the light flux in an area having a low light flux on the shade.

[0157] On the other hand, the vehicle lamp **110** is provided with the single first lens portion **143**, the single second lens portion **144**, and the single third lens portion **145** that collect light from the respective three light sources (**124**, **125**, **126**) in the collecting lens **113**. The first lens portion **143** forms, on the shade portion **133** of the shade **114**, the first light distribution area in which the vicinity of the apex of the first slit portion **1361** is the brightest while the entire area of the first slit portion **1361** is brightened with the light from the corresponding first light source **124**. In addition, the second lens portion **144** forms, on the shade portion **133** of the shade **114**, the second light distribution area AL in which the vicinity of the apex of the second slit portion **1362** is the brightest while the entire area of the second slit portion **1362** is brightened with the light from the corresponding second light source **125**. The third lens portion **145** forms, on the shade portion **133** of the shade **114**, the third light distribution area to brighten the entire area of the third slit portion **1363** with a uniform light flux lower than that of the first light distribution area and the second light distribution area AL.

[0158] Here, the third slit portion **1363** corresponds to the third radiation design Di3 located closest to the vehicle **101** in the radiation pattern Pi and is projected onto a place close to the vehicle lamp **110**, and therefore, in order to form the third radiation design Di3 which is easy to see, it is desirable that the light flux be uniform and lower than that of the other radiation designs Di. This is due to the following reasons. Since the third slit portion **1363** corresponds to the third radiation design Di3, the third slit portion **1363** has a large dimension on the shade portion **133**. Then, when the third slit portion **1363** is irradiated by using a lens having a predetermined light flux distribution in which the light flux is modulated as in the first lens portion **143** or the second lens portion **144**, it is difficult to irradiate the entire area, and unevenness of the brightness, such as the occurrence of a dark place, occurs in the formed third radiation design Di3, and there is a concern that the appropriate third radiation design Di3 cannot be formed. On the other hand, since the first slit portion **1361** and the second slit portion **1362** correspond to the first radiation design Di1 and the second radiation design Di2 at positions away from the vehicle **101** in the radiation pattern Pi and are projected onto places distant from the vehicle lamp **110**, it is desirable to have a light flux distribution with a predetermined modulation (inflection (differences in the level of the light flux)) in order to form the first radiation design Di1 and the second radiation design Di2 which are easy to see.

[0159] The vehicle lamp **110** can illuminate the shade **114** with a desired light distribution by providing the first lens portion **143**, the second lens portion **144**, and the third lens portion **145** individually corresponding to the respective light sources (**124**, **125**, **126**) in the single collecting lens **113**. That is, the vehicle lamp **110** forms the first light distribution area and the second light distribution area AL having a light flux distribution with a predetermined modulation by using the first lens portion **143** and the second lens portion **144** for the first slit portion **1361** and the second slit portion **1362** and forms the third light distribution area having a uniform light flux by using the third lens portion **145** for the third slit portion **1363**. Therefore, even with the use of the single collecting lens **113**, the vehicle lamp **110** can form both a portion having a predetermined light flux distribution with modulation and a portion having a uniform light flux on the shade portion **133**. As a result, the vehicle lamp **110** can make the middle of the leading end of the first radiation design Di1 and the second radiation design Di2 clear as a center, can make the entire third radiation design Di3 clear, and can form the more appropriate radiation pattern Pi. Therefore, as compared with conventional vehicle lamps, the vehicle lamp **110** can easily adjust the light distribution of the light distribution area formed on the shade **114** and can form the radiation pattern Pi that has a desired brightness distribution and is easy to see, while having a simple configuration using the single collecting lens **113**.

[0160] Here, in the vehicle lamp **110**, in order to individually correspond to each of the light sources (**124**, **125**, **126**) by the single collecting lens **113**, the first lens portion **143**, the second lens portion **144**, and the third lens portion **145** having different optical characteristics are overlapped in

the up-down direction and integrally formed in the collecting lens main body **131** of the collecting lens **113**. Then, in the vehicle lamp **110**, a part of the light from each light source (**124**, **125**, **126**) travels to a lens portion (**143**, **144**, **145**) different from the corresponding one, and an unintended light distribution area may be formed on the shade portion **133** of the shade **114** due to the light. This will be described below.

[0161] First, since the collecting lens main body **131** of the collecting lens **113** is configured as described above, basically, the light from the first light source **124** travels to the first lens portion **143** (the first opposing incidence surface portion **151** and the first inclined incidence surface portion **152** thereof), the light from the second light source **125** travels to the second lens portion **144** (the second opposing incidence surface portion **157** and the second inclined incidence surface portion **158** thereof), and the light from the third light source **126** travels to the third lens portion **145** (the third incidence surface **164** thereof). However, since each of the light sources (**124**, **125**, **126**) has a predetermined spread in the direction in which the light is emitted, a part of the light may be directed to a lens portion adjacent to the corresponding lens portion. This example is illustrated in FIGS. **20** and **21**. In FIG. **20**, the collecting lens **113** (the collecting lens main body **131**) having the same configuration as that of the present invention is used. This is because the same problem occurs even in the collecting lens **113** (the collecting lens main body **131**) when the first light reduction portion **171** and the second light reduction portion **172** are not provided. For this reason, in the following description, the description will be given using unintended light that may occur in a case where the front-surface-side first light reduction portion **173** and the front-surface-side second light reduction portion **175** of the front surface **141** and the back-surface-side first light reduction portion **174** and the back-surface-side second light reduction portion **176** of the back surface **142** are not provided in the collecting lens **113** (the collecting lens main body **131**).

[0162] FIG. **20** illustrates an example of the situation where a part of the light (hereinafter, also referred to as stray light S) from the second light source **125** travels toward the shade **114** through the portion (the portion corresponding to the second light reduction portion **172**) between the second lens portion **144** and the third lens portion **145**. The stray light S is incident on the collecting lens main body **131** from between the second opposing incidence surface portion **157** or the second inclined incidence surface portion **158** of the second lens portion **144** and the third incidence surface **164** of the third lens portion **145** (defined as stray light S1). The stray light S travels between the second lens portion **144** and the third lens portion **145** (the portion corresponding to the second light reduction portion **172**) (defined as stray light S2), is emitted from between the second emission surface **161** of the second lens portion **144** and the third emission surface **165** of the third lens portion **145**, and travels toward the third slit portion **1363** of the shade **114** (defined as stray light S3). FIG. **21** illustrates a light distribution on the shade portion **133** of the shade **114** when only the second light source **125** is turned on and the above-described stray light S occurs.

[0163] In the shade portion **133**, as described above, the light from the second light source **125** is emitted through the second lens portion **144** so that the second light distribution area AL is formed, in which the vicinity of the apex of the second slit portion **1362** is the brightest while the entire area of the second slit portion **1362** is brightened. When the above-described stray light S has occurred, the stray light S is directed toward the third slit portion **1363** of the shade **114**, and thus an unintended light distribution area (hereinafter, referred to as a stray light area AS) is formed in the third slit portion **1363**.

[0164] The stray light area AS brightens the vicinity of the center of the third slit portion **1363** even though the third light source **126** is turned off. Therefore, when the stray light S occurs in the vehicle lamp **110**, even in a case where the second light source **125** is turned on while the third light source **126** is turned off, the second radiation design Di2 can be clearly formed, but a part of the area where the third radiation design Di3 is formed is brightened in a blurred manner. In addition, when the stray light S occurs in the vehicle lamp **110**, even in a case where the second

light source **125** and the third light source **126** are simultaneously turned on, an unintended area of the third radiation design Di3 is brightened corresponding to the stray light area AS, brightness unevenness occurs, and the appearance is different from the intended appearance, although the second radiation design Di2 can be appropriately formed.

[0165] On the other hand, in the vehicle lamp **110**, the front-surface-side second light reduction portion **175** and the back-surface-side second light reduction portion **176** as the second light reduction portion **172** are provided between the second lens portion **144** and the third lens portion **145** in the collecting lens main body **131**. Therefore, the above-described stray light S1 is blocked or scattered by the back-surface-side second light reduction portion **176** on the back surface **142** of the collecting lens main body **131** and is prevented from traveling into the collecting lens main body **131** like the stray light S2. That is, the back-surface-side second light reduction portion **176** can block the incidence of the stray light S1 to prevent the stray light S1 from becoming the stray light S2 or can diffuse the stray light S1 to prevent the stray light S2, which is strong light (high light flux) forming the stray light area AS, from traveling into the collecting lens main body **131**. Further, after passing through the back-surface-side second light reduction portion **176**, the stray light S2 is blocked or scattered by the front-surface-side second light reduction portion **175** on the front surface **141** of the collecting lens main body **131**. Thus, the stray light S2 is blocked or diffused by the front-surface-side second light reduction portion **175** after being diffused and sufficiently weakened by the back-surface-side second light reduction portion **176** as described above, and therefore, it is possible to prevent the stray light S3, which is strong light (high light flux) forming the stray light area AS, from being emitted from the collecting lens main body **131** in a more reliable manner.

[0166] Further, although the above-described stray light S2 may occur even when the stray light does not pass through the back-surface-side second light reduction portion **176**, the stray light S2 is blocked or scattered by the front-surface-side second light reduction portion **175** on the front surface **141** of the collecting lens main body **131**, and is prevented from being emitted from the collecting lens main body **131** like the stray light S3. Accordingly, the vehicle lamp **110** can prevent the stray light area AS from being formed in the third slit portion **1363** of the shade **114** due to the stray light S from the second light source **125** as described above. Therefore, when the vehicle lamp **110** turns off the third light source **126** and turns on the second light source **125**, the second radiation design Di2 can be clearly formed, and the area where the third radiation design Di3 is formed can be kept dark. In addition, when the second light source **125** and the third light source **126** are turned on at the same time, the vehicle lamp **110** can appropriately form the second radiation design Di2 and the third radiation design Di3, and can obtain an intended appearance.

[0167] Here, the stray light as described above may cause the same problem even when the light from the third light source **126** travels from between the second lens portion **144** and the third lens portion **145** toward the second slit portion **1362** of the shade **114**. On the other hand, the vehicle lamp **110** can also block or scatter such stray light by the front-surface-side second light reduction portion **175** and the back-surface-side second light reduction portion **176** as the second light reduction portion **172**.

[0168] Further, the stray light described above may cause the same problem even when the light from the second light source **125** travels from between the second lens portion **144** and the first lens portion **143** toward the first slit portion **1361** of the shade **114** or when the light from the first light source **124** travels from between the second lens portion **144** and the first lens portion **143** toward the second slit portion **1362** of the shade **114**. In contrast, in the vehicle lamp **110**, since the first light reduction portion **171** is provided between the first lens portion **143** and the second lens portion **144** in the collecting lens main body **131**, such stray light can also be blocked or scattered by the front-surface-side first light reduction portion **173** and the back-surface-side first light reduction portion **174**. For these reasons, the vehicle lamp **110** can appropriately form each of the radiation designs Di, and can make each of the radiation designs Di look as intended.

[0169] Here, the collecting lens main body **131** (the collecting lens **113**) according to the second embodiment includes, as the first light reduction portion **171** and the second light reduction portion **172**, the front-surface-side first light reduction portion **173** and the front-surface-side second light reduction portion **175** of the front surface **141** and the back-surface-side first light reduction portion **174** and the back-surface-side second light reduction portion **176** of the back surface **142**.

Therefore, the first light reduction portion **171** and the second light reduction portion **172** cannot block or scatter the light traveling between the lens portions (**143**, **144**, **145**) in the collecting lens main body **131**. However, since the collecting lens main body **131** is provided with the three lens portions (**143**, **144**, **145**) individually corresponding to the three light sources (**124**, **125**, **126**), it is possible to suppress the light incident on the corresponding lens portion (**143**, **144**, **145**) from each light source (**124**, **125**, **126**) from being directed to the adjacent lens portion. In particular, in the collecting lens main body **131** according to the second embodiment, each of the lens portions (**143**, **144**, **145**) causes the light from the corresponding light source (**124**, **125**, **126**) to be incident as parallel light traveling in a substantially parallel manner, and therefore, the amount of light traveling toward an adjacent lens portion can be significantly reduced. For this reason, even when the collecting lens main body **131** (the collecting lens **113**) includes the first light reduction portion **171** and the second light reduction portion **172** formed by performing the light reduction processing on predetermined positions of the front surface **141** and the back surface **142**, the influence of the stray light described above can be sufficiently suppressed.

[0170] In addition, the collecting lens main body **131** (the collecting lens **113**) according to the second embodiment is provided with the bottom-surface-side second light reduction portion **177** forming a part of the second light reduction portion **172** by performing the light reduction processing also on the bottom surface **145a** located on the second lens portion **144** side in the third lens portion **145**. For this reason, in the third lens portion **145**, it is possible to suppress the light from being incident from the bottom surface **145a** opposed to the second light source **125** and becoming stray light due to the difference in shape between the third incidence surface **164** and the second opposing incidence surface portion **157**, the second inclined incidence surface portion **158**, and the second reflective surface **159** in the second lens portion **144**. Therefore, when the second light source **125** is turned on while the third light source **126** is turned off, the collecting lens main body **131** (the collecting lens **113**) can keep the area where the third radiation design Di3 is formed dark, and when the second light source **125** and the third light source **126** are turned on at the same time, the second radiation design Di2 and the third radiation design Di3 can be appropriately formed.

[0171] The vehicle lamp **110** according to the second embodiment can obtain the following effects.

[0172] The vehicle lamp **110** includes the plurality of light sources (**124**, **125**, **126**), the collecting lens **113** that collects light from them, the shade **114** provided with the plurality of slit portions **136** that partially passes the collected light, and the projecting lens **115** that projects the light having passed therethrough to form the radiation pattern Pi including the plurality of radiation designs Di corresponding to the plurality of slit portions **136**. The light sources (**124**, **125**, **126**) are provided individually corresponding to the slit portions **136**, and the collecting lens **113** includes the plurality of lens portions (**143**, **144**, **145**) that individually corresponds to the slit portions **136** and overlaps with each other and is provided with the light reduction portions (**171**, **172**) that reduces light between the plurality of lens portions (**143**, **144**, **145**). Therefore, the vehicle lamp **110** can sufficiently suppress the influence of the stray light in which the light incident on the corresponding lens portion (**143**, **144**, **145**) from each light source (**124**, **125**, **126**) is directed to the slit portion **136** different from the corresponding one.

[0173] In the vehicle lamp **110**, the light reduction portions (**171**, **172**) are formed by performing light reduction processing between the plurality of lens portions (**143**, **144**, **145**) on the back surface **142** on the plurality of light sources (**124**, **125**, **126**) side in the collecting lens **113**.

Therefore, the vehicle lamp **110** can suppress the brightness of the place of the radiation design Di

which is not turned on or can suppress the occurrence of uneven brightness of the formed radiation design Di and thus the appearance different from the intended one due to the light (the stray light S1.fwdarw.S2) which is incident on the collecting lens 113 from between the lens portions (143, 144, 145).

[0174] In the vehicle lamp 110, the light reduction portions (171, 172) are formed by performing light reduction processing between the plurality of lens portions (143, 144, 145) on the surface 141 that is the shielding member (the shade 114) side in the collecting lens 113. Therefore, the vehicle lamp 110 can suppress the brightness of the place of the radiation design Di which is not turned on or can suppress the occurrence of uneven brightness of the formed radiation design Di and thus the appearance different from the intended one due to the light (the stray light S2.fwdarw.S3) emitted from between the lens portions (143, 144, 145).

[0175] The vehicle lamp 110 can sequentially turn on the plurality of light sources (124, 125, 126). Therefore, since the vehicle lamp 110 can prevent the place of the radiation design Di corresponding to the turned-off light source from being brightened, it is possible to appropriately express the intention of sequentially turning on the light sources.

[0176] In the vehicle lamp 110, the plurality of slit portions 136 includes the first slit portion 1361 corresponding to the first radiation design Di1 of the radiation pattern Pi, the second slit portion 1362 corresponding to the second radiation design Di2 of the radiation pattern Pi, and the third slit portion 1363 corresponding to the third radiation design Di3 of the radiation pattern Pi. Further, the plurality of lens portions (143, 144, 145) includes the first lens portion 143 opposed to the first slit portion 1361, the second lens portion 144 opposed to the second slit portion 1362, and the third lens portion 145 opposed to the third slit portion 1363. The first lens portion 143 includes the first opposing incidence surface portion 151 opposed to the corresponding first light source 124, the first inclined incidence surface portion 152 surrounding it, and the first reflective surface 153 surrounding it. In addition, the second lens portion 144 includes the second opposing incidence surface portion 157 opposed to the corresponding second light source 125, the second inclined incidence surface portion 158 surrounding it, and the second reflective surface 159 surrounding it, and the third lens portion 145 is a convex lens that collects light from the corresponding third light source 126. Therefore, the vehicle lamp 110 can efficiently use the light from the first light source 124 and the second light source 125, and can form a predetermined light flux distribution in the first slit portion 1361 and the second slit portion 1362 while simplifying the configuration of the first lens portion 143 and the second lens portion 144. In addition, the vehicle lamp 110 can form a uniform light flux distribution by using the third lens portion 145 for the third slit portion 1363 with the light from the third light source 126. For these reasons, even with the use of the single collecting lens 113, the vehicle lamp 110 can make the middle of the leading ends of the first radiation design Di1 and the second radiation design Di2 clear as a center, can make the entire third radiation design Di3 clear, and can form the more appropriate radiation pattern Pi.

[0177] In the vehicle lamp 110, in the third lens portion 145, the light reduction processing is also performed on the bottom surface 145a located on the second lens portion 144 side. Therefore, when the second light source 125 is turned on while the third light source 126 is turned off, the vehicle lamp 110 can keep the area where the third radiation design Di3 is formed dark, and when the second light source 125 and the third light source 126 are turned on at the same time, the second radiation design Di2 and the third radiation design Di3 can be appropriately formed.

[0178] In the vehicle lamp 110, the plurality of lens portions (143, 144, 145) is integrated. Therefore, in the vehicle lamp 110, it is possible to increase the relative positional accuracy between the plurality of lens portions (143, 144, 145) and to facilitate the assembly process and the like.

[0179] Therefore, the vehicle lamp 110 according to the second embodiment as the vehicle lamp according to the present disclosure can form the radiation pattern Pi having a desired brightness distribution while efficiently using the light from the plurality of light sources (124, 125, 126).

[0180] Although the vehicle lamp according to the present disclosure has been described above based on the second embodiment, the specific configuration is not limited to the second embodiment, and design changes, additions, and the like are allowed without departing from the gist of the invention according to each claim in the scope of claims.

[0181] According to the second embodiment, the radiation pattern **Pi** is configured such that the three radiation designs **Di** have a substantially isosceles triangle shape having a base on the vehicle **101** side and are arranged at substantially equal intervals in a direction away from the vehicle **101**. However, as long as the radiation pattern includes the plurality of radiation designs **Di** formed by the shade (shielding member), the design of the symbol as the radiation design **Di**, the formation position, the number of the radiation designs **Di**, and the like, may be appropriately set, and are not limited to the configuration according to the second embodiment.

[0182] Although the vehicle lamp **110** is provided in the front portion of the vehicle **101** according to the second embodiment, the vehicle lamp **110** may be accommodated in a door mirror, may be provided in a lamp chamber of a headlight or a lamp chamber of a tail lamp (lamp chambers on both right and left sides of a rear portion of the vehicle), or may be provided in a vehicle body as long as it is provided in the vehicle **101** in accordance with the position where the radiation pattern is formed for the vehicle **101**, and is not limited to the configuration according to the second embodiment.

[0183] Furthermore, according to the second embodiment, each of the light sources (**124**, **125**, **126**) emits amber light. However, the color of the light emitted from the light source may be appropriately set in accordance with the arrangement location and the content to be conveyed, and is not limited to the configuration according to the second embodiment.

[0184] According to the second embodiment, the shade **114** that causes the light collected by the collecting lens **113** to pass through the radiating slit **135** is used as the shielding member. However, the shielding member may have another configuration as long as it is provided with the plurality of slit portions **136** (the radiating slits **135**) that partially passes the light collected by the collecting lens **113**, and is not limited to the configuration according to the second embodiment. As another configuration, for example, it is possible to use a shielding plate (filter) in which a plurality of radiating slits that partially transmits light is provided in a plate-shaped film member that blocks the transmission of light and the light having passed through the collecting lens **113** is transmitted through the plurality of radiating slits.

[0185] According to the second embodiment, the vehicle lamp **110** is provided in the vehicle **101** driven by a driver. However, the vehicle lamp may be provided in a vehicle having an automatic driving function, and is not limited to the configuration according to the second embodiment. In this case, the vehicle lamp may form the radiation pattern at the timing corresponding to the usage for which the vehicle lamp is provided, i.e., at the timing corresponding to some intention regarding the operation of the vehicle **101**, and is not limited to the configuration according to the second embodiment.

[0186] According to the second embodiment, the light source portion **112** is provided on the installation base portion **111** having a function as a heat sink, and the collecting lens **113**, the shade **114**, and the projecting lens **115** are attached to the installation base portion **111**. However, the vehicle lamp may have a different configuration as long as the light from the light source is collected on the shielding member by the collecting lens and is projected by the projecting lens to form the radiation pattern, and is not limited to the configuration according to the second embodiment.

[0187] According to the second embodiment, each of the light sources (**124**, **125**, **126**) includes an LED chip and a phosphor covering it. However, as long as the light sources (**124**, **125**, **126**) individually correspond to the lens portions (**143**, **144**, **145**), the configuration of each of the light sources may be appropriately set and is not limited to the configuration according to the second embodiment.

[0188] According to the second embodiment, first, the third light source **126** is turned on, then the second light source **125** is turned on while the third light source **126** is continuously turned on, then the first light source **124** is turned on while the third light source **126** and the second light source **125** are continuously turned on, then the light sources (**124, 125, 126**) are turned off at the same time, and thereafter this is repeated. However, all of the light sources (**124, 125, 126**) may be turned on at the same time, all of the light sources may be turned on individually, or the light sources may be turned on in combination at random as appropriate, the order in which the light sources are turned on and the mode in which the light sources are turned on can be set as appropriate, and are not limited to the mode according to the second embodiment.

[0189] According to the second embodiment, the first light reduction portion **171** and the second light reduction portion **172** include the front-surface-side first light reduction portion **173** and the front-surface-side second light reduction portion **175** of the front surface **141** and the back-surface-side first light reduction portion **174** and the back-surface-side second light reduction portion **176** of the back surface **142** in the collecting lens main body **131** (the collecting lens **113**). However, each of the light reduction portions (**171, 172**) is not limited to the configuration according to the second embodiment as long as the light reduction portion is provided between the corresponding lens portions (**143, 144, 145**) to block or scatter the progress of light. For example, each of the light reduction portions (**171, 172**) may be formed in a plate shape between the corresponding lens portions (**143, 144, 145**), i.e., may block or scatter the light traveling between the lens portions (**143, 144, 145**) in the collecting lens main body **131**.

[0190] Hereinafter, a vehicle lamp according to a third embodiment of the present disclosure will be described with reference to the drawings. It should be noted that the present invention is not limited to the third embodiment. In addition, constituent elements according to the third embodiment described below include those that can be easily replaced by those skilled in the art or those that are substantially the same.

Third Embodiment

[0191] FIG. **22** is an exploded perspective view illustrating an example of a vehicle lamp **200** according to the third embodiment. FIG. **23** is a diagram illustrating a state of the vehicle lamp **200** when viewed from the front. FIG. **24** is a diagram illustrating a configuration along a cross-section of F-F in FIG. **23**. FIG. **25** is a diagram illustrating a configuration along a cross-section of G-G in FIG. **23**. FIG. **25** is an enlarged view of a portion fixed to a fixing portion **253**. As illustrated in FIGS. **22** to **25**, the vehicle lamp **200** includes a light source portion **210**, a collecting lens member **220**, a shielding member **230**, a projecting lens member **240**, and a supporting member **250**.

[0192] The light source portion **210** includes a light source **211** and a substrate **212**. The light source **211** is, for example, a semiconductor-type light source such as an LED. The light source **211** includes a light emission surface **211a** that emits light. The light emission surface **211a** is provided to be opposed to a collecting lens **221** of the collecting lens member **220**. The light source **211** emits, for example, orange (amber) light from the light emission surface **211a**. The light sources **211** are arranged side by side in the up-down direction in a vehicle-mounted state. For example, the two light sources **211** are provided. The arrangement and the number of the light sources **211** are not limited to those described above. Further, the color of the light emitted from the light emission surface **211a** is not limited to orange, and may be another color such as white.

[0193] The light source **211** is mounted on the substrate **212**. A wiring, a circuit, and the like for transmitting a signal to the light source **211** are formed on the substrate **212**. The substrate **212** is fixed to a base portion **251** of the supporting member **250** described below.

[0194] The collecting lens member **220** includes a collecting lens **221**, a cylindrical portion **222**, and a collecting lens frame **223**. In the collecting lens member **220**, the collecting lens **221**, the cylindrical portion **222**, and the collecting lens frame **223** are formed as one member. The entire collecting lens member **220** is formed using the material forming the collecting lens **221**. The collecting lens **221** is formed using a material capable of transmitting light from the light source

211. Examples of such a material include a resin material such as polycarbonate, but other materials such as acrylic may be used. In this case, the whole of the collecting lens member **220** can be easily formed by integral molding using the material forming the collecting lens **221**. In the collecting lens member **220**, a portion different from the collecting lens **221**, i.e., at least a part of the cylindrical portion **222** and the collecting lens frame **223** may be formed using a material different from that of the collecting lens **221**.

[0195] The collecting lens **221** collects the light emitted from the light source **211**. As illustrated in FIG. **24**, the collecting lens **221** includes a first incidence surface **221a**, a second incidence surface **221b**, a reflective surface **221c**, and an emission surface **221d**. The first incidence surface **221a** is provided in front of the light source **211**. The light emitted forward from the light source **211** is incident on the first incidence surface **221a**.

[0196] The second incidence surface **221b** is provided along the outer periphery of the first incidence surface **221a**. The light emitted from the light source **211** in an inclined manner with respect to an optical axis AX is incident on the second incidence surface **221b**. According to the third embodiment, an optical axis direction D is a direction along the optical axis AX. According to the third embodiment, the optical axis direction D coincides with the front-back direction.

[0197] The reflective surface **221c** internally reflects the light incident from the second incidence surface **221b**.

[0198] The emission surface **221d** emits forward the light incident from the first incidence surface **221a** and the light incident from the second incidence surface **221b** and internally reflected by the reflective surface **221c**.

[0199] The cylindrical portion **222** holds the collecting lens **221**. The cylindrical portion **222** has, for example, a cylindrical shape. The cylindrical portion **222** connects the collecting lens **221** and the collecting lens frame **223**. The cylindrical portion **222** is provided to protrude backward (toward the light source) with respect to the collecting lens frame **223**. With this configuration, the collecting lens **221** is provided behind the collecting lens frame **223**.

[0200] The collecting lens frame **223** holds the collecting lens **221** via the cylindrical portion **222**. The collecting lens frame **223** has a flat plate shape. The collecting lens frame **223** includes an annular portion **225** protruding in the up-down direction from the cylindrical portion **222** and a band-shaped portion **226** protruding in the right-left direction from the cylindrical portion **222**. The annular portion **225** is provided annularly along the outer periphery of the cylindrical portion **222**. The band-shaped portion **226** is provided linearly in the right-left direction from the cylindrical portion **222**. Corner portions on both sides in the right-left direction of the band-shaped portion **226** have a rounded shape. The band-shaped portion **226** is supported by an end surface **253c** of the fixing portion **253** of the supporting member **250** described below. The band-shaped portion **226** includes a positioning opening **226a** and a fixing opening **226b**. A positioning protrusion **253a** of the supporting member **250** described below is inserted into the positioning opening **226a**. A fixing member **260** described below is inserted into the fixing opening **226b**. The band-shaped portion **226** includes contact portions **226c** and **226d**. The contact portion **226c** is provided along the outer periphery of the positioning opening **226a** and protrudes forward from the band-shaped portion **226**. The contact portion **226d** is provided along the outer periphery of the fixing opening **226b** and protrudes forward from the band-shaped portion **226**. The contact portions **226c**, **226d** are formed so that the leading end surface in the protruding direction is flat. The collecting lens frame **223** is in contact with the shielding member **230** at the contact portions **226c**, **226d**. With this configuration, it is possible to secure the positional accuracy of the collecting lens frame **223** by appropriately defining the dimension such as the height in the protruding direction of the contact portions **226c**, **226d** of the band-shaped portion **226**.

[0201] The shielding member **230** includes a slit formation portion **231** and a shielding frame **232**. In the shielding member **230**, the slit formation portion **231** and the shielding frame **232** are formed in a flat plate shape as one member. The entire shielding member **230** is formed using a material

capable of shielding light. Examples of such a material include a material such as a metal, but other materials may be used.

[0202] The slit formation portion **231** is provided in, for example, a circular shape. The slit formation portion **231** includes a slit **233**. The slit **233** causes a part of the light collected by the collecting lens **221** to pass therethrough. For example, the slits **233** are formed in a state where three slits are arranged in the up-down direction. The number and arrangement of the slits **233** are not limited to those described above.

[0203] The shielding frame **232** protrudes linearly in the right-left direction from the slit formation portion **231**. Corner portions on both sides in the right-left direction of the shielding frame **232** have a rounded shape. Both front and back surfaces of the shielding frame **232** are flat. The shielding frame **232** is supported by the contact portions **226c**, **226d** of the band-shaped portion **226** of the collecting lens frame **223**. Since the shielding frame **232** is supported by the contact portions **226c**, **226d**, the shielding frame **232** includes a positioning opening **232a** and a fixing opening **232b**. The positioning protrusion **253a** of the supporting member **250** described below is inserted into the positioning opening **232a**. The fixing member **260** described below is inserted into the fixing opening **232b**.

[0204] The shielding member **230** is formed to cover the collecting lens **221** and the collecting lens frame **223** of the collecting lens member **220** when viewed from the front. For example, the slit formation portion **231** is provided at the position corresponding to the collecting lens **221** and is formed to have the shape and the dimension corresponding to the collecting lens **221**. In addition, the shielding frame **232** is provided at the position corresponding to the collecting lens frame **223** and is formed to have the shape and the dimension corresponding to the collecting lens frame **223**. According to the third embodiment, the shielding member **230** is formed to have the same or substantially the same outer shape as the collecting lens member **220** when viewed from the front. With this configuration, the light having passed through the collecting lens **221** and the collecting lens frame **223** can be blocked by the shielding member **230**.

[0205] The projecting lens member **240** includes a projecting lens **241**, a cylindrical portion **242**, and a projecting lens frame **243**. The projecting lens **241** projects the light having passed through the slit **233** onto the road surface in front of the vehicle to form a radiation pattern. In the projecting lens member **240**, the projecting lens **241**, the cylindrical portion **242**, and the projecting lens frame **243** are formed as one member. The entire projecting lens member **240** is formed using a material that transmits light. The projecting lens **241** is formed using a material capable of transmitting light from the light source **211**. Examples of such a material include a resin material such as acrylic, but other materials may be used. In this case, the entire projecting lens member **240** can be easily formed by integral molding using the material forming the projecting lens **241**. The entire projecting lens member **240** is formed of a material different from that of the collecting lens member **220**, but may be entirely formed of the same material as that of the collecting lens member **220**. In the projecting lens member **240**, a portion different from the projecting lens **241**, i.e., at least a part of the cylindrical portion **242** and the projecting lens frame **243** may be formed using a material different from that of the projecting lens **241**.

[0206] As illustrated in FIG. **24**, the projecting lens **241** includes an incidence surface **241a** and an emission surface **241b**. The light having passed through the slit **233** is incident on the incidence surface **241a**. The emission surface **241b** emits forward the light incident from the incidence surface **241a**.

[0207] The cylindrical portion **242** holds the projecting lens **241**. The cylindrical portion **242** has, for example, a cylindrical shape. The cylindrical portion **242** connects the projecting lens **241** and the projecting lens frame **243**. The cylindrical portion **242** is provided to protrude forward with respect to the projecting lens frame **243**. With this configuration, the projecting lens **241** is provided in front of the projecting lens frame **243**.

[0208] The projecting lens frame **243** holds the projecting lens **241** via the cylindrical portion **242**.

The projecting lens frame **243** has a flat plate shape. The projecting lens frame **243** includes an annular portion **245** that protrudes from the cylindrical portion **242** in the up-down direction and a band-shaped portion **246** that protrudes from the cylindrical portion **242** in the right-left direction. The annular portion **245** is provided annularly along the outer periphery of the cylindrical portion **242**. The band-shaped portion **246** is provided linearly in the right-left direction from the cylindrical portion **242**. Corner portions on both sides in the right-left direction of the band-shaped portion **246** have a rounded shape. The band-shaped portion **246** includes a positioning opening **246a** and a fixing opening **246b**. The positioning protrusion **253a** of the supporting member **250** described below is inserted into the positioning opening **246a**. The fixing member **260** described below is inserted into the fixing opening **246b**. The band-shaped portion **246** includes contact portions **246c**, **246d**. The contact portion **246c** is provided along the outer periphery of the positioning opening **246a** and protrudes backward from the band-shaped portion **246**. The contact portion **246d** is provided along the outer periphery of the fixing opening **246b** and protrudes backward from the band-shaped portion **246**. The contact portions **246c**, **246d** are formed so that the leading end surface in the protruding direction is flat. The projecting lens frame **243** is in contact with the shielding member **230** at the contact portions **246c**, **246d**. With this configuration, it is possible to ensure the positional accuracy of the projecting lens frame **243** by appropriately defining the dimension, such as the height in the protruding direction, of the contact portions **246c**, **246d** of the band-shaped portion **246**.

[0209] The supporting member **250** includes a base portion **251**, a fin **252**, and a fixing portion **253**. The base portion **251** has a flat plate shape. The base portion **251** includes a support surface **251a** that supports the light source portion **210**. The support surface **251a** is a front surface of the base portion **251** and supports the substrate **212**.

[0210] The fin **252** protrudes backward from the base portion **251**. The plurality of fins **252** is provided. The fins **252** radiate the heat generated in the light source **211**.

[0211] The fixing portion **253** protrudes forward from the support surface **251a** of the base portion **251**. The fixing portion **253** fixes the collecting lens frame **223**, the shielding frame **232**, and the projecting lens frame **243**. The fixing portion **253** includes a positioning protrusion **253a** and a fixing opening **253b**. The positioning protrusion **253a** and the fixing opening **253b** are provided on a front end surface **253c** of the fixing portion **253**.

[0212] The positioning protrusion **253a** protrudes forward and penetrates through the positioning opening **226a** provided in the band-shaped portion **226** of the collecting lens frame **223**, the positioning opening **232a** provided in the shielding frame **232**, and the positioning opening **246a** provided in the band-shaped portion **246** of the projecting lens frame **243** in the front-back direction. The fixing member **260** described below is inserted into the fixing opening **253b**.

[0213] The end surface **253c** is formed, for example, in a flat shape. As illustrated in FIG. 25, the end surface **253c** is formed to be perpendicular or substantially perpendicular to the optical axis AX. The collecting lens member **220** is supported by the end surface **253c** so that the collecting lens member **220**, the shielding member **230**, and the projecting lens member **240** overlap with each other in the optical axis direction D. That is, the end surface **253c** serves as a reference surface for supporting the collecting lens member **220**, the shielding member **230**, and the projecting lens member **240**. Since the end surface **253c** serving as a reference surface is formed to be perpendicular or substantially perpendicular to the optical axis AX, the collecting lens member **220**, the shielding member **230**, and the projecting lens member **240** can be appropriately provided along the optical axis AX.

[0214] The fixing member **260** fixes the collecting lens frame **223**, the shielding frame **232**, and the projecting lens frame **243** to the fixing portion **253**. As the fixing member **260**, for example, a fastening member such as a screw is used. The fixing member **260** is inserted into the fixing opening **253b** of the fixing portion **253** by penetrating through the fixing opening **226b** provided in the band-shaped portion **226** of the collecting lens frame **223**, the fixing opening **232b** provided in

the shielding frame **232**, and the fixing opening **246b** provided in the band-shaped portion **246** of the projecting lens frame **243**.

[0215] Next, an example of an operation of the vehicle lamp **200** configured as described above will be described. FIG. **26** is a diagram illustrating an example of an operation of the vehicle lamp **200**. When the driver performs a predetermined operation such as an operation of a vehicle-side direction indicator or an operation of a hazard switch, the vehicle lamp **200** emits light from the light emission surfaces **211a** of the light source **211** in response to the operation.

[0216] A part of light L emitted from the light emission surface **211a** is incident on the first incidence surface **221a** and the second incidence surface **221b** of the collecting lens **221**. Light L1 incident on the first incidence surface **221a** travels inside the collecting lens **221** and is emitted forward from the emission surface **221d**. Light L2 incident on the second incidence surface **221b** is reflected forward by the reflective surface **221c** and is emitted forward from the emission surface **221d**. Further, a part of the light L emitted from the light emission surface **211a** enters the cylindrical portion **222** or the collecting lens frame **223**. Light L3 incident on the cylindrical portion **222** or the collecting lens frame **223** is transmitted through the cylindrical portion **222** and the collecting lens frame **223** and is emitted forward.

[0217] The lights L1, L2 emitted forward from the emission surface **221d** and the light L3 emitted forward from the cylindrical portion **222** or the collecting lens frame **223** reach the slit formation portion **231** of the shielding member **230**. A part of the lights L1, L2 having reached the slit formation portion **231** passes through the slit **233** of the shielding member **230**, and the remaining light is shielded by the shielding member **230**. Further, the light L3 reaches the shielding frame **232** of the shielding member **230**, and is shielded by the shielding frame **232**. According to the third embodiment, the collecting lens member **220** and the shielding member **230** are provided such that their outer shapes overlap with each other when viewed from the optical axis direction D.

Therefore, the light L3 emitted forward from a portion other than the collecting lens **221** in the collecting lens member **220**, i.e., the cylindrical portion **222** or the collecting lens frame **223**, is shielded by the shielding member **230**. Therefore, the occurrence of glare can be prevented.

[0218] The lights L1, L2 having passed through the slits **233** are incident on the incidence surface **241a** of the projecting lens **241** and are emitted from the emission surface **241b** to the front of the vehicle. FIG. **27** is a diagram illustrating an example of a radiation pattern formed on a road surface by the vehicle lamp **200**. As illustrated in FIG. **27**, a radiation pattern P is formed on the road surface ahead of the vehicle with the lights L1, L2 emitted to the front of the vehicle.

[0219] As described above, the vehicle lamp **200** according to the third embodiment includes the light source **211**, the collecting lens member **220** in which the collecting lens **221** that collects light emitted from the light source **211** and the collecting lens frame **223** holding the collecting lens **221** are formed as one member, the shielding member **230** in which the slit formation portion **231** including the slit **233** that partially passes a part of the light collected by the collecting lens **221** and the shielding frame **232** holding the slit formation portion **231** are formed as one member, the projecting lens member **240** provided with the projecting lens **241** that projects the light having passed through the slit **233** onto the road surface to form the radiation pattern P, and the supporting member **250** that supports the light source **211**, and the collecting lens member **220**, the shielding member **230**, and the projecting lens member **240** are fixed to the supporting member **250** such that the collecting lens frame **223** and the shielding frame **232** overlap with each other in the optical axis direction D.

[0220] With this configuration, since the collecting lens frame **223** and the shielding frame **232** are fixed to the supporting member **250** to overlap with each other in the optical axis direction D, it is possible to reduce the number of components required for fixing each optical member of the collecting lens member **220**, the shielding member **230**, and the projecting lens member **240**. By reducing the number of parts required for fixing, each optical member can be accurately attached.

[0221] In the vehicle lamp **200** according to the third embodiment, the entire collecting lens

member **220** is formed using a material that transmits light, and the shielding member **230** is formed to cover the collecting lens **221** and the collecting lens frame **223** of the collecting lens member **220** when viewed from the front. With this configuration, it is possible to form the entire collecting lens member **220** by using the material forming the lens portion that transmits light. Therefore, the collecting lens member **220** can be easily formed at low costs. Further, in this configuration, even when the light from the light source **211** is transmitted through a portion other than the collecting lens **221** of the collecting lens member **220**, the light can be shielded by the shielding member **230**. Therefore, the occurrence of glare light can be prevented.

[0222] In the vehicle lamp **200** according to the third embodiment, the projecting lens **241** and the projecting lens frame **243** holding the projecting lens **241** are formed as one member in the projecting lens member **240**, and the collecting lens member **220**, the shielding member **230**, and the projecting lens member **240** are fixed to the supporting member **250** such that the collecting lens frame **223**, the shielding frame **232**, and the projecting lens frame **243** overlap with each other in the optical axis direction D. With this configuration, in addition to the collecting lens frame **223** and the shielding frame **232**, the projecting lens frame **243** is also fixed to the supporting member **250** to overlap in the optical axis direction D, and thus it is possible to accurately attach each optical member.

[0223] In the vehicle lamp **200** according to the third embodiment, the collecting lens **221** is provided behind the collecting lens frame **223** in the optical axis direction in the collecting lens member **220**, and the projecting lens **241** is provided in front of the projecting lens frame **243** in the optical axis direction in the projecting lens member **240**. With this configuration, the collecting lens **221** and the projecting lens **241** can be efficiently arranged.

[0224] In the vehicle lamp **200** according to the third embodiment, the supporting member **250** includes the base portion **251** including the support surface **251a** that supports the light source **211** and the fixing portion **253** that fixes the collecting lens frame **223**, the shielding frame **232**, and the projecting lens frame **243**, and the fixing portion **253** protrudes from the support surface **251a**. With this configuration, it is possible to sufficiently secure a space for arranging the collecting lens **221** in the supporting member **250**.

[0225] In the vehicle lamp **200** according to the third embodiment, the fixing portion **253** is provided with the end surface **253c** serving as a reference surface to fix the collecting lens frame **223**, the shielding frame **232**, and the projecting lens frame **243** at the leading end in the protruding direction, and the end surface **253c** is formed to be orthogonal to the optical axis AX. With this configuration, it is possible to appropriately arrange the collecting lens member **220**, the shielding member **230**, and the projecting lens member **240** along the optical axis AX.

[0226] The technical scope of the present invention is not limited to the above-described third embodiment, and modifications can be made without departing from the spirit of the present invention. For example, according to the third embodiment described above, the configuration in which the entire collecting lens member **220** is formed using a material that transmits light has been described as an example, but the present invention is not limited thereto. The collecting lens member **220** may have a configuration in which at least the collecting lens **221** is formed using a material that transmits light, and the cylindrical portion **222** and the collecting lens frame **223** may be formed using a material that does not transmit light.

[0227] Similarly, according to the third embodiment described above, the configuration in which the entire projecting lens member **240** is formed using a material that transmits light has been described as an example, but the present invention is not limited thereto. The projecting lens member **240** may have a configuration in which at least the projecting lens **241** is formed using a material that transmits light, and the cylindrical portion **242** and the projecting lens frame **243** may be formed using a material that does not transmit light.

[0228] According to the third embodiment described above, the configuration in which the collecting lens **221** is provided behind the collecting lens frame **223** in the optical axis direction in

the collecting lens member **220** and the projecting lens **241** is provided in front of the projecting lens frame **243** in the optical axis direction in the projecting lens member **240** has been described as an example, but the present invention is not limited thereto. For example, the thicknesses of the collecting lens frame **223** and the projecting lens frame **243** may be the same as those of the collecting lens **221** and the projecting lens **241**, respectively.

[0229] In addition, according to the third embodiment, the configuration in which the fixing portion **253** protrudes from the support surface **251a** in the supporting member **250** has been described as an example, but the present invention is not limited thereto. For example, the collecting lens frame **223** may be configured to protrude toward the support surface **251a**.

[0230] According to the third embodiment described above, the configuration in which the vehicle lamp **200** is provided at the front portion of a vehicle M has been described as an example, but the present invention is not limited thereto. The vehicle lamp **200** may be configured to be provided on a rear portion or a side portion of the vehicle M, and may be configured to form a radiation pattern on a road surface behind or beside the vehicle M.

DESCRIPTION OF REFERENCE NUMERALS

[0231] **10** VEHICLE LAMP [0232] **14** COLLECTING LENS [0233] **15** SHADE (AS EXAMPLE OF SHIELDING MEMBER) [0234] **16** PROJECTING LENS [0235] **31** FIRST LIGHT SOURCE [0236] **32** SECOND LIGHT SOURCE [0237] **53** RADIATING SLIT [0238] **541** FIRST SLIT PORTION (AS EXAMPLE OF FAR-SIDE SLIT PORTION) [0239] **542** SECOND SLIT PORTION (AS EXAMPLE OF FAR-SIDE SLIT PORTION) [0240] **543** THIRD SLIT PORTION (AS EXAMPLE OF NEAR-SIDE SLIT PORTION) [0241] **61** FIRST LENS PORTION [0242] **62** SECOND LENS PORTION [0243] **65** CURVED INCIDENCE SURFACE PORTION [0244] **66** ANNULAR INCIDENCE SURFACE PORTION [0245] **67** REFLECTIVE SURFACE [0246] **110** VEHICLE LAMP [0247] **113** COLLECTING LENS [0248] **114** SHADE (AS EXAMPLE OF SHIELDING MEMBER) [0249] **115** PROJECTING LENS [0250] **124** FIRST LIGHT SOURCE [0251] **125** SECOND LIGHT SOURCE [0252] **126** THIRD LIGHT SOURCE [0253] **136** SLIT PORTION [0254] **1361** FIRST SLIT PORTION [0255] **1362** SECOND SLIT PORTION [0256] **1363** THIRD SLIT PORTION [0257] **141** FRONT SURFACE [0258] **142** BACK SURFACE [0259] **143** FIRST LENS PORTION [0260] **144** SECOND LENS PORTION [0261] **145** THIRD LENS PORTION [0262] **145a** BOTTOM SURFACE [0263] **151** FIRST OPPOSING INCIDENCE SURFACE PORTION [0264] **152** FIRST INCLINED INCIDENCE SURFACE PORTION [0265] **153** FIRST REFLECTIVE SURFACE [0266] **157** SECOND OPPOSING INCIDENCE SURFACE PORTION [0267] **158** SECOND INCLINED INCIDENCE SURFACE PORTION [0268] **159** SECOND REFLECTIVE SURFACE [0269] **171** FIRST LIGHT REDUCTION PORTION (AS EXAMPLE OF LIGHT REDUCTION PORTION) [0270] **172** SECOND LIGHT REDUCTION PORTION (AS EXAMPLE OF LIGHT REDUCTION PORTION) [0271] **200** VEHICLE LAMP [0272] **210** LIGHT SOURCE PORTION [0273] **211** LIGHT SOURCE [0274] **211a** LIGHT EMISSION SURFACE [0275] **212** SUBSTRATE [0276] **220** COLLECTING LENS MEMBER [0277] **221** COLLECTING LENS [0278] **221a** FIRST INCIDENCE SURFACE [0279] **221b** SECOND INCIDENCE SURFACE [0280] **221c** REFLECTIVE SURFACE [0281] **221d**, **241b** EMISSION SURFACE [0282] **222**, **242** CYLINDRICAL PORTION [0283] **223** COLLECTING LENS FRAME [0284] **225**, **245** ANNULAR PORTION [0285] **226**, **246** BAND-SHAPED PORTION [0286] **226a**, **232a**, **246a** POSITIONING OPENING [0287] **226b**, **232b**, **246b**, **253b** FIXING OPENING [0288] **226c**, **226d**, **246c**, **246d** CONTACT PORTION [0289] **230** SHIELDING MEMBER [0290] **231** SLIT FORMATION PORTION [0291] **232** SHIELDING FRAME [0292] **233** SLIT [0293] **240** PROJECTING LENS MEMBER [0294] **241** PROJECTING LENS [0295] **241a** INCIDENCE SURFACE [0296] **243** PROJECTING LENS FRAME [0297] **250** SUPPORTING MEMBER [0298] **251** BASE PORTION [0299] **251a** SUPPORT SURFACE [0300] **252** FIN [0301] **253** FIXING PORTION [0302] **253a** POSITIONING PROTRUSION [0303] **253c** END SURFACE [0304] **260** FIXING MEMBER [0305] **Ai** INNER RING LIGHT

DISTRIBUTION AREA [0306] A_o OUTER RING LIGHT DISTRIBUTION AREA [0307] A₁ FIRST LIGHT DISTRIBUTION AREA [0308] A₂ SECOND LIGHT DISTRIBUTION AREA [0309] A_X OPTICAL AXIS [0310] D OPTICAL AXIS DIRECTION [0311] D_i RADIATION DESIGN [0312] D_{i1} FIRST RADIATION DESIGN [0313] D_{i2} SECOND RADIATION DESIGN [0314] D_{i3} THIRD RADIATION DESIGN [0315] L, L₁, L₂, L₃ LIGHT [0316] L_p PROJECTION OPTICAL AXIS [0317] M VEHICLE [0318] P_i RADIATION PATTERN

Claims

1. A vehicle lamp comprising: a first light source and a second light source arranged in parallel; a collecting lens that collects light from the first light source and the second light source; a shielding member that is provided with a radiating slit that partially passes the light collected by the collecting lens; and a projecting lens that projects the light having passed through the shielding member to form a radiation pattern, wherein the collecting lens includes a first lens portion that corresponds to the first light source and a second lens portion that corresponds to the second light source, the radiating slit includes a near-side slit portion that corresponds to a near-side radiation design projected onto a near position in the radiation pattern and a far-side slit portion that corresponds to a far-side radiation design projected onto a farther position than the near-side radiation design in the radiation pattern, the first lens portion is provided to be opposed to the far-side slit portion, and the second lens portion is provided to be opposed to the near-side slit portion.
2. The vehicle lamp according to claim 1, wherein the first lens portion forms, on the shielding member, a first light distribution area to irradiate an entire area of the far-side slit portion with light from the first light source, and the second lens portion forms, on the shielding member, a second light distribution area to irradiate an entire area of the near-side slit portion with light from the second light source.
3. The vehicle lamp according to claim 2, wherein the first lens portion includes a curved incidence surface portion opposed to the first light source, an annular incidence surface portion surrounding the curved incidence surface portion, and a reflective surface surrounding the annular incidence surface portion, the second lens portion is a convex lens that collects light from the second light source, the first lens portion forms, on the shielding member, an inner ring light distribution area to irradiate the far-side slit portion with light from the first light source via the curved incidence surface portion and forms an outer ring light distribution area to irradiate the entire far-side slit portion while providing a light flux in the vicinity of a center of the far-side slit portion higher than surroundings with light from the first light source having passed through the annular incidence surface portion and reflected by the reflective surface, and the first light distribution area is formed such that the inner ring light distribution area and the outer ring light distribution area overlap with each other.
4. The vehicle lamp according to claim 3, wherein the second light distribution area has a lower light flux than the inner ring light distribution area and the outer ring light distribution area and has a smaller light flux difference than the outer ring light distribution area.
5. The vehicle lamp according to claim 1, wherein the first lens portion and the second lens portion are integrated.
6. The vehicle lamp according to claim 1, wherein, in the shielding member, three quarters or more of the far-side slit portion are provided below a projection optical axis of the projecting lens.
7. The vehicle lamp according to claim 1, wherein the first light source and the second light source are arranged in parallel in a vertical direction, and the first light source is located below the second light source.
8. The vehicle lamp according to claim 2, wherein the far-side radiation design includes a first radiation design on a far side and a second radiation design closer to the near-side radiation design than the first radiation design, the far-side slit portion includes a first slit portion that corresponds

to the first radiation design and a second slit portion that corresponds to the second radiation design, and the first light distribution area provides, in the vicinity of a center of the first slit portion, the highest light flux in the first light distribution area and the second light distribution area.

9. A vehicle lamp comprising: a plurality of light sources; a collecting lens that collects light from the plurality of light sources; a shielding member that is provided with a plurality of slit portions that partially passes the light collected by the collecting lens; and a projecting lens that projects the light having passed through the shielding member to form a radiation pattern including a plurality of radiation designs that corresponds to the plurality of slit portions, wherein the light sources are provided individually corresponding to the slit portions, and the collecting lens includes a plurality of lens portions that individually corresponds to the slit portions and overlaps with each other, and the collecting lens is provided with a light reduction portion that reduces light between the plurality of lens portions.

10. The vehicle lamp according to claim 9, wherein the light reduction portion is formed by performing light reduction processing between the plurality of lens portions on a back surface that is a side of the plurality of light sources in the collecting lens.

11. The vehicle lamp according to claim 9, wherein the light reduction portion is formed by performing light reduction processing between the plurality of lens portions on a front surface that is a side of the shielding member in the collecting lens.

12. The vehicle lamp according to claim 9, wherein the plurality of light sources is sequentially turned on.

13. The vehicle lamp according to claim 9, wherein the plurality of slit portions includes a first slit portion that corresponds to a first radiation design projected onto a distant position in the radiation pattern, a second slit portion that corresponds to a second radiation design projected onto a closer position than the first radiation design in the radiation pattern, and a third slit portion that corresponds to a third radiation design projected onto a closer position than the second radiation design in the radiation pattern, the plurality of lens portions includes a first lens portion opposed to the first slit portion, a second lens portion opposed to the second slit portion, and a third lens portion opposed to the third slit portion, the first lens portion includes a first opposing incidence surface portion opposed to the corresponding light source, a first inclined incidence surface portion surrounding the first opposing incidence surface portion, and a first reflective surface surrounding the first inclined incidence surface portion, the second lens portion includes a second opposing incidence surface portion opposed to the corresponding light source, a second inclined incidence surface portion surrounding the second opposing incidence surface portion, and a second reflective surface surrounding the second inclined incidence surface portion, and the third lens portion is a convex lens that collects light from the corresponding light source.

14. The vehicle lamp according to claim 13, wherein, in the third lens portion, light reduction processing is also performed on a bottom surface located on the second lens portion side.

15. The vehicle lamp according to claim 9, wherein the plurality of lens portions is integrated.

16. A vehicle lamp comprising: a light source; a collecting lens member in which a collecting lens that collects light emitted from the light source and a collecting lens frame holding the collecting lens are formed as one member; a shielding member in which a slit formation portion including a slit that partially passes a part of the light collected by the collecting lens and a shielding frame holding the slit formation portion are formed as one member; a projecting lens member provided with a projecting lens that projects the light having passed through the slit onto a road surface to form a radiation pattern; and a supporting member that supports the light source, wherein the collecting lens member, the shielding member, and the projecting lens member are fixed to the supporting member such that the collecting lens frame and the shielding frame overlap with each other in an optical axis direction.

17. The vehicle lamp according to claim 16, wherein the entire collecting lens member is formed

using a material forming the collecting lens, and the shielding member is formed to cover the collecting lens and the collecting lens frame of the collecting lens member when viewed from the front.

18. The vehicle lamp according to claim 16, wherein in the projecting lens member, the projecting lens and a projecting lens frame holding the projecting lens are formed as one member, and the collecting lens member, the shielding member, and the projecting lens member are fixed to the supporting member such that the collecting lens frame, the shielding frame, and the projecting lens frame overlap with each other in the optical axis direction.

19. The vehicle lamp according to claim 18, wherein the collecting lens of the collecting lens member is provided behind the collecting lens frame in the optical axis direction, and the projecting lens of the projecting lens member is provided in front of the projecting lens frame in the optical axis direction.

20. The vehicle lamp according to claim 19, wherein the supporting member includes a base portion including a support surface that supports the light source and a fixing portion that fixes the collecting lens frame, the shielding frame, and the projecting lens frame, and the fixing portion protrudes from the support surface.

21. The vehicle lamp according to claim 20, wherein the fixing portion is provided with an end surface serving as a reference surface to fix the collecting lens frame, the shielding frame, and the projecting lens frame at a leading end in a protruding direction, and the end surface is formed to be orthogonal to the optical axis direction.
